Graduation Project (GP)

Football Market Analysis

Dissertation

Information System (IS)

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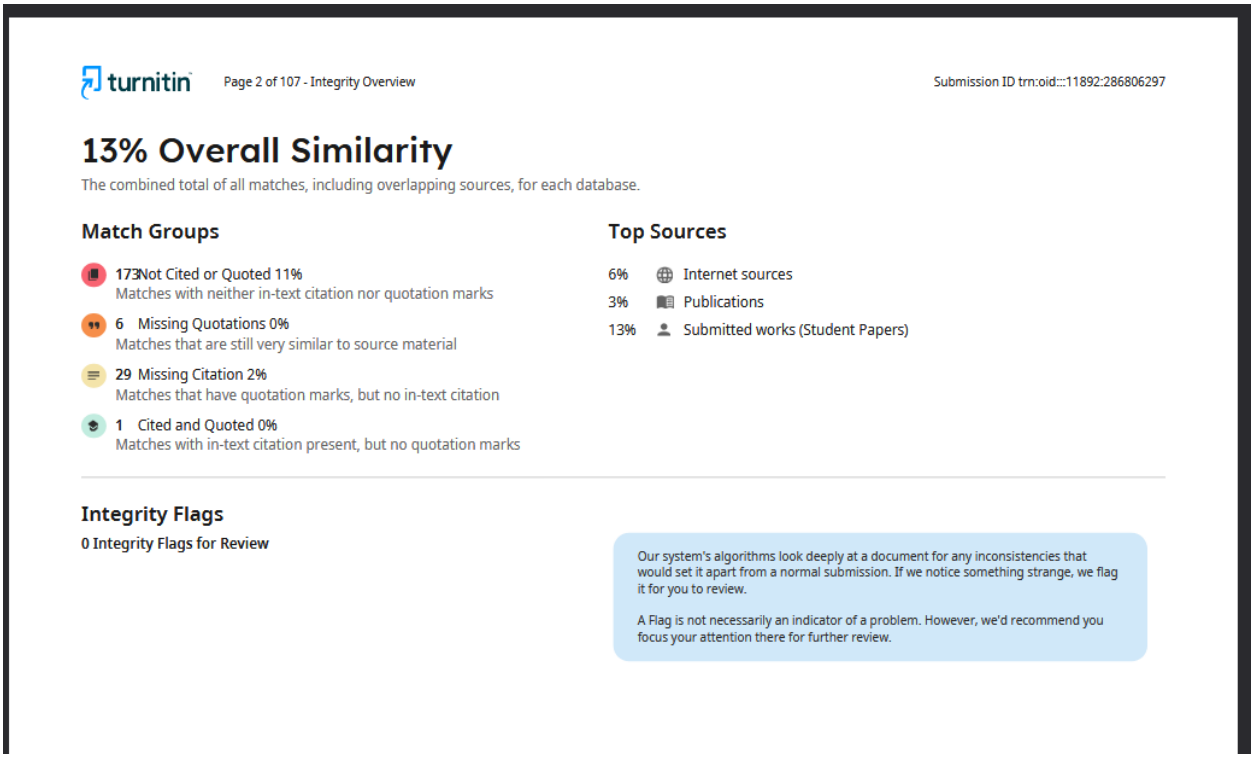
Abstract

The football transfer market has experienced unprecedented inflation over the past decade, with clubs often overpaying for players due to hype and speculation rather than objective performance indicators. This project addresses this issue by building a machine learning-based system that predicts football players' market value based on real-world data from Transfermarkt and Football Manager. The solution integrates multiple models, including an ensemble of Linear Regression and Random Forest Tree, as well as an optimized XGBoost model, to capture both linear and complex non-linear patterns influencing player valuations.

Data preprocessing techniques, feature engineering, and performance evaluations were carried out to ensure reliable predictions. The project also involved developing a full-stack web application using Vue.js for the frontend and Node.js with MongoDB for the backend. The system includes additional functionalities such as a player value generator, customizable squad builder, and match analysis tools. Evaluation results showed a significant improvement compared to recreated academic models, demonstrating the effectiveness of advanced feature creation and model blending. This work provides a comprehensive, scalable tool for football clubs, analysts, and fans to make smarter, data-driven decisions in the transfer market.

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Introduction

The introduction sets the stage for the project by outlining its purpose and motivation.
It presents the background, problem statement, scope, and main objectives of the study.
This section also summarizes the structure of the report and the approach followed.

1.1 Overview

Football is the world's most-watched sport, leading to high profits for the European football clubs According to Deloitte the total revenue generated by the top 20 clubs in 2022/23 was a record €10.5bn [3]. The most import decisions for football clubs are to choose players who fit the team's dynamics and to pay an accurate price that fits with the player's talent, potential, and previous performances. García-del-Barrio and Rossi (2023) have shown that incorrect player valuation can cause significant financial losses and destabilize club management strategies [1].

Since the start of the century, the amount of money exchanged hands between clubs has been almost dizzying According to the CIES Football Observatory's September 2024 report, the total global investment in player transfer fees reached €10.96 billion in 2024, marking the second-highest level recorded after 2023, which reached €12.24 billion [4], while some of these transfers have been worth it or even a bargain others have been a trainwreck for both the player and the club. Research by Müller et al. (2021) show that inflated transfer prices unrelated to a player's actual on-field contribution results from speculative market behavior [5].

In this regard, precisely estimating a player's market value becomes even more crucial than it ever has. Data-driven models are being used more and more by football teams, agents, and analysts to evaluate players' present value and future potential even for salary negotiations. Comparatively to conventional assessment techniques, recent developments in machine learning—especially ensemble approaches like XGBoost—have shown great performance in estimating football player market values [2].

With an eye toward supporting more informed decision-making in the football transfer market, this dissertation investigates the use of machine learning and data mining approaches to predict football player market values. By means of multidimensional player data—including technical skills, market history, and performance measures—Azevedo et al. (2023) can greatly enhance the fairness and transparency of valuations, so benefiting buying and selling clubs [6].

The global football industry has evolved into a multi-billion-euro enterprise, where clubs make strategic and financial decisions that can significantly impact their sporting success and commercial viability. One of the most critical aspects of this business is the transfer market, where clubs buy and sell players based on perceived value, tactical fit, and potential return on investment. However, in recent years, player transfer fees have soared to record levels, often influenced by speculation, media hype, and market inflation—leading to questionable deals and

overspending. A 2023 systematic review by García-del-Barrio and Rossi emphasized that market values are frequently distorted by non-performance-related factors such as media narratives and public sentiment [1]. Additionally, research by Lozano et al. (2023) introduced a debiased machine learning approach that demonstrated how subjective biases can be significantly reduced when evaluating player performances [2].

This part describes the project, which intends to solve this problem by means of a machine learning-based model based on real-world and simulation-based data, so precisely estimating the market value of football players. Recent studies, including those by Azevedo et al. (2023), have shown that Random Forest and XGboost, among other ensemble machine learning methods, significantly outperform conventional statistical methods in projecting player valuations [3]. Moreover, the following sections give background on the project by means of a thorough review of the present situation of football transfers, a clear problem statement, and the extent and goals of the suggested solution. The work method clarifies the research and decision-making process that molded the design of the model; the report structure is meant to help the reader traverse the material. At last, a Gantt chart shows the deliverables and schedule of the project.

1.2 Problem Statement

Europe clubs now compete with each other in the transfer window, leading to an all-time high in the transfer fees paid for players, which causes overpaying. As a result, clubs need to be smart with their money and find a way to evaluate the player value subjectively, free from media hype or market speculation.

Europe clubs now compete with each other in the transfer window, leading to an all-time high in the transfer fees paid for players, which causes overpaying. As a result, clubs need to be smart with their money and find a way to evaluate the player value subjectively, free from media hype or market speculation.

This issue is clearly illustrated in recent transfer activity. According to Statista [7], several clubs in the 2023 summer transfer window paid significantly more than the estimated market value of players. For example, Chelsea paid €116 million for Moisés Caicedo, while his assessed value was only €76.4 million a disparity of nearly €40 million. Similarly, Arsenal paid €116.6 million for Declan Rice, with a value estimate of just €77.3 million. Other cases such as Neymar (€90M fee vs. €51.7M value) and Rúben Neves (€55M fee vs. €32.4M value) further highlight the widening gap between fee and value which is even more clear in the below Figure.

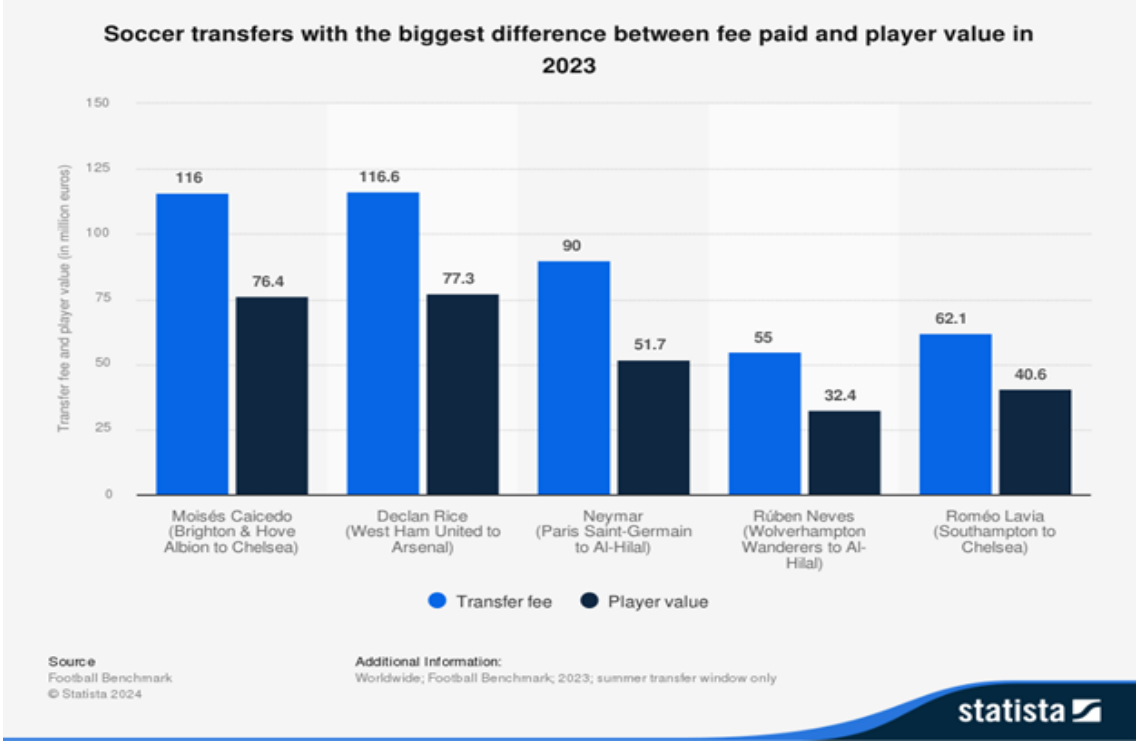


Figure 1 Difference between Fee paid and player vale 2023

The challenge, then, is to create a model capable of assessing a player's influence and importance, translating these factors into an accurate market valuation free from external influence. This valuation can then serve as a foundation for smarter transfer decisions, optimized budget allocations, and more strategic squad-building practices.

1.3 Scope and Objectives

In this section the scope and objectives will be discussed for the project.

1.3.1 Scope

To develop a model that accurately estimates football players' market value, it is essential to use reliable data sources. such as Transfermarkt, which offers real-world statistics including player contributions such as goals, assists, and clean sheets, as well as additional information like contract expiration dates and club affiliations. These data points are crucial for assessing a player's impact and role within a team.

1.3.2 Objectives

The primary objective of this project is to build a machine learning model that predicts football players' market values using data aggregated from multiple sources. The model aims to eliminate subjectivity and bias by relying solely on data-driven feature analysis.

This tool will assist buying clubs in determining whether they are paying a fair price, and provide selling clubs with evidence to justify their asking prices. Additionally, the model can support contract negotiations by offering an unbiased valuation of a player based on performance metrics.

Overall, the goal is to enhance transparency and fairness in the football transfer market through intelligent, data-backed decision-making.

1.4 Report Organization

This paper is structured into nine main parts to present the project clearly and efficiently. **Part 1, Introduction**, outlines the background, problem statement, goals, and overall timeline. **Part 2, Related Work**, reviews previous studies and highlights how this project offers a distinct contribution. **Part 3, Methodology**, explains the workflow of the project. **Part 4, Implementation**, details the development process. **Part 5, Testing and Evaluation**, evaluates model performance. **Part 6, Results and Discussion**, presents a comparison of model outcomes. **Part 7, Diagrams and Design**, includes system diagrams and web application screenshots. **Part 8, Conclusions and Changes**, summarizes the key results and updates made during the project. Finally, **Part 9, Future Work**, proposes potential improvements and future enhancements

1.5 Work Methodology

- I first looked at several studies aimed at developing models able to project football players' transfer market value.
- I picked 11 papers related to the player market value prediction using machine learning.
- I made a comparison table covering the main ideas of every paper together with their objectives, methods, datasets, and limitations.
- Out of the 11 papers, I chose three most pertinent ones that closely matched the model I meant to create.
- I repeated reading the selected three papers several times to fully grasp their methods.
- I hand drew basic graphs to show the flow of every chosen paper and improved understanding of their models.
- This process guided my choice of the best algorithms for my project by helping me to grasp the advantages and drawbacks of various techniques.

1.6 Work Plan (Gantt Chart)

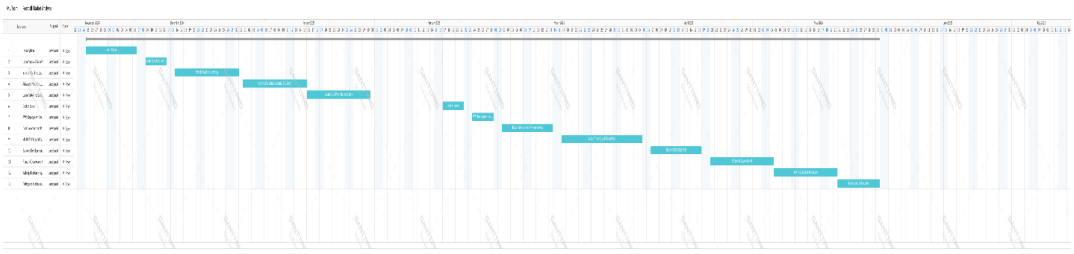


Figure 2 Gnatt chart

2. Related Work

In this part I will provide a more background for my project and discuss pervious research paper that covers the same topic.

2.1 Background

According to the statistical model from the International Center for Sports Studies (CIES), multiple factors that impacts a player's transfer market value. These variables can be categorized into three main categories: **Player Characteristics**, **Performance Metrics**, **Team and Competition Context** [3].

In the following sections, we will explore each category in detail to understand how these factors affecting the player market value.

1) Player Characteristics

The age of players and the duration of their contracts are the most important factors in estimating a player's transfer market value. Typically, younger players have a greater value in the transfer market because of their longer career, development potential, and higher resale value, which makes them a more appealing long-term investments for clubs. The length of a player's contract with their current club is also an important factor in the negotiations for a player's sale.as players whose contracts expire can join another club as free agents without any fees required. Therefore, this can be used during negotiations as a pressure point for the selling club into lowering the asking price for a player's transfer. On the other hand, players with long-term contracts usually require higher transfer fees, as the selling club retains more control over the player's future in negotiations.

2) **Performance Metrics**

Performance Metrics such as playing time and performance level have a role. The amount of play time for each player points to the importance of the player to the team. Also, the performance level is important as it indicates the player's quality such as participation in top-tier leagues or international competitions, reflects a player's experience and skill level.

3) **Team and Competition Context**

The context of the team and competition has a major impact on a player's market value, especially through the financial power of both the purchasing and selling clubs. Clubs with strong financial resources are capable of asking for higher transfer fees when selling players and are also able to pay higher prices when acquiring new talent. On the other hand, clubs with limited financial means may need to sell players for less to address urgent financial obligations, which diminishes their negotiating strength. Besides, teams from richer leagues, such as the English Premier League, frequently outbid those from weaker leagues finically, which further affects how players are valued.

2.2 Literature Survey

McHale and Holmes (2022) [4] developed models to estimate football player transfer fees using a dataset comprising player statistics, match actions, and transfer histories. They employed four algorithms, starting with Linear Regression (OLS) as a baseline to capture simple relationships, achieving a Mean Absolute Error (MAE) of 10.34, Mean Absolute Percentage Error (MAPE) of

91.81%, and R^2 of 0.50. Elastic-Net Regression (glmnet) was then used to select influential features among correlated data, which improved the MAE to 9.93, with a similar MAPE and R^2 of 0.50. The XGBoost Tree model proved to be the best-performing algorithm, effectively handling complex data patterns with an MAE of 3.60, MAPE of 67.47%, and a significant R^2 of 0.77. Finally, the authors introduced a variation called XGboost with Dropout (xgbDART), which achieved an MAE of 3.64, MAPE of 68.90%, and R^2 of 0.76 by preventing overfitting through the random dropping of trees. XGBoost tree provide the most accuracy capturing 77% accuracy.

Nikolaos(2024) [5]explored factors influencing football player transfer fees through different datasets including transfer datasets, FIFA datasets, and real-life statistics datasets Several machine learning were used to make the model such as Decision Trees which was used to capture non-linear patterns as the baseline, which resulted in an R^2 of 0.33 with a log transformation and a Root Mean Square Error (RMSE) of 0.83. Next, K-Nearest Neighbors (KNN) was used to improve predictions by identifying similar players based on their attributes, achieving an R^2 of 0.45 and an RMSE of 0.72. LightGBM, a gradient-boosting algorithm, was applied for improved computational efficiency and accuracy, yielding an R^2 of 0.68 and RMSE of 0.57. Finally, XGBoost, recognized for its efficiency and scalability in capturing complex patterns, demonstrated the best performance with an R^2 of 0.70 and RMSE of 0.55. The study concluded that XGBoost, with optimized hyperparameters and regularization, outperformed other models in terms of prediction accuracy.

Al-Asadi and Tasdemir [6] investigated the market transfer value of football players datasets from FIFA 20 and sofifa.com as well, which includes over 17,000 players. The study used four machine learning models to predict player value, beginning with Linear Regression as a baseline model, which achieved an RMSE of 5.46 and an R^2 of 0.43. Multiple Linear Regression was then employed, adding more features to enhance prediction accuracy, with an improved RMSE of 4.66 and R^2 of 0.56. Regression trees were used to capture non-linear relationships resulting in an RMSE of 2.71 and ^{R2} of 0.87. The Random Forest tree proved to be the most effective in reducing overfitting and delivering the highest predictive accuracy with an RMSE of 1.64 and R^2 of 0.95. Al-Asadi and Tasdemir reached the conclusion that Random Forest was the most suited for player market value prediction due to its ability to handle complex relationships.

Franceschi et al. [7] performed a systematic review of football players' valuation by analyzing 29 empirical studies. The datasets included actual transfer fees, crowdsourced values from platforms such as Transfermarkt, and media-derived valuations from sources like Kicker and Marca. Franceschi et al used several algorithms such as OLS Regression and the Generalized Method of Moments (GMM).

Sulimov (2024) [8] developed a model to predict football players' transfer fees, adding statistical metrics such as expected goals (xG). The model used data From StatsBomb and football data from Kaggle, and employed algorithms such as CatBoost and Random Forest tree. The research concluded that CatBoost was the most effective model.

Lombardi et al. (2020) [9] researched the link between football players' skills to the transfer market value. The research used bivariate linear regression and network analysis and used datasets from Transfermarkt to analyze the transfer market activity between 2016/17 and 2017/18 seasons. The

conclusion reached was that the transfer market value of players is affected by serval factors like player's ability to improve, the clubs' interested in the player, and other economic factors.

Podzemský (2022) [10] researched the relationship between football clubs' activity and their successes. The model was made using algorithms such as Pearson correlation and Ordinary Least Squares (OLS) regression to examine the relationship or not. The study focused on Europe's top five leagues and displayed that Sera A the highest correlation coefficient (0.87).

Alaminos et al. (2023) [11] used advanced neural network methodologies such as Multilayer Perceptron (DRCNN), Deep Neural Decision Trees (DNDT), and Quantum Neural Networks (QNN). The model used data from Eurostat and STOXX Europe Football Index (2011–2020) as the source of data. To make a model able to estimate football clubs listed on the stock exchange. The study shows that advanced algorithms provide more accurate estimation than simple algorithms such as Ordinary Least Square (OLS).

Behravan and Razavi (2020) [12] designed a model that is capable of predicting football player market value using data from FIFA 20. The model employs APSO-clustering to group players into four clusters using algorithms such as the support Vector Regression (SVR) model for prediction. The model achieved 74% accuracy.

DiBlasi (2022) [13] developed a valuation model for footballers based on their transfer fees and subsequent season performance in Europe's top five leagues. Using data from Transfermarkt and WhoScored, the study included variables such as player age, popularity, performance metrics (e.g., goals, assists, and clean sheets), and positional characteristics.

2.3 Analysis

The following table contains comparison between 11 different research papers that discusses model that is cable of estimating the player's market valuation

Ref	Contribution	Algorithms	Datasets	Limitations
[4] McHale and Holmes (2022)	Estimated player market cost by analyzing player performance and applying machine-learning models for transfer value predictions.	• Linear Regression. • Ordinary Least Squares (OLS) Linear Regression. • Mixed Effects Linear Model (glmer)	• Transfer Data • Performance Metrics • Crowd-Sourced Ratings • Club Financial Data	• checking whether the prices paid for players were reasonable based on their performance after the transfer.

[5] Nikolaos' (2024)	Use data from FIFA and real-life metrics to predict transfer fees, helping clubs in adhering to Financial Fair Play regulations.	• Baseline Model (Decision Trees). • K-nearest neighbors (KNN). • XGBoost. • LightGBM	• FIFA Dataset. • Transfers Dataset • Real-Life Statistics Dataset	• Incorporating more statistical metrics to improve the model's accuracy in predicting transfer fees • Exploring the prediction of market values or transfer fees for female players, considering the increasing popularity of women's football
[6] Al-Asadi and Tasdemir's (2022)	Developed a model using FIFA data to assist clubs and agents in more accurate player valuations during negotiations	• BASELINE MODEL (LINEAR REGRESSION) • MULTIPLE LINEAR REGRESSION • REGRESSION TREE • RANDOM FOREST REGRESSION	• FIFA 20 Datasets. • Transfermarkt Data. • sofifa datasets	• Try to build a calculator to aid in valuing the player transfer cost
[7] (Gouguet, 2024)	Conduct a systematic review to identify key drivers of football players' market valuation.	• Generalized Method of Moments (GMM) • Quantile OLS Regression. • Feasible Generalized Least Squares (FGLS).	• Transfermarkt Data. • Media Valuations e.g. (MARCA)	• The salary was not given significant role in player valuation, future studies could include salary as it's a major factor

		• Tobit Models. • Two-Stage Least Squares (2SLS).		
[8] (Sulimov 2024)	Design an AI-driven model to predict the transfer fees of football players. The aim of the model is to be used by football clubs to make better recruitment decision	• Logistic Regression. • Random Forest Classifier/Regressor. • CatBoost Classifier/Regressor. • Decision Tree Regressor	• StatsBomb Open Data. • Kaggle Football Data.	• Including additional features like the remaining contract term, the economic level of the recruiting club, and a number of injuries could improve the model's predictive capabilities.
[9] (Lombardi et al., 2020)	Investigated knowledge transfer between clubs, emphasizing key factors for successful knowledge exchange.	• Bivariate Linear Regression Analysis. • Network Analysis. • UCINET Software.	• Transfermarkt Data. • Network Analysis Data. • Performance Data.	• Investigate additional football seasons and different leagues to provide a broader understanding of knowledge transfer.
[10] (Polák & Červinka, 2022)	Investigate the relationship between the market value of a team and their success in European football.	• Correlation Analysis. • Ordinary Least Squares. • Variance Inflation Factor. • Breusch-Pagan.	• Transfermarkt Data. • Stadium Attendance. • League Data. • UEFA Rankings.	• **No Limitations mentioned.**

[11] (Alaminos et al. ,2023)	Make an asset pricing model for football clubs to provide a more accurate estimation of value of football club shares. To represent a better evaluation of the football clubs assets.	• Multilayer Perceptron (DRCNN). • Deep Neural Decision Trees (DNDT). • Quantum Neural Networks (QNN)	• Eurostat and Stoxx Europe Football Index. • Bureau Van Dijk Amadeus database. • Sample of 21 football clubs.	• The authors suggest expanding the model to include data from football clubs in other regions . • The study is limited to old STOXX Europe Football from 2011-2019.
[12] (Behravan & Razavi, 2020)	create a model that can help evaluate players with better accuracy that Transfermarkt	• APSO-Clustering. • PSO-SVR.	• FIFA 20 Dataset. • Transfermarkt.com Data	• Improving the accuracy of the player value estimation model beyond the current 74%. • Incorporating more recent datasets or additional features to refine the model.
[13] (DiBlasi, n.d.)	Create a model cable of estimating value of football players by comparing their transfer fees and comparing value to their performances.	• Ordinary Least Squares (OLS) Regression • Lasso Model	• Transfermakrt • WhoScored • Google Trends • The Guardian	exploring the impact of international soccer on players' transfer values and examining different positions within a position group.

[14] (Coates & Parshakov, 2021)	To check whether available public data can be used to improve the ability of predicting transfer fee of players.	• Ordinary Least Squares (OLS) Regression • Quantile Regression • Principal Components Analysis • Hypothesis Testing	• Transfermakrt • EA FIFA	• incorporating a wider array of player skills and characteristics to provide more insight • Modeling the effect of time remaining on the contract as dependent upon player quality could refine the analysis

Research Papers Comparison

- Most research papers use Transfermarkt and FIFA datasets, while my project uniquely integrates Transfermarkt and Football Manager for deeper player insights.
- My goal is to create a web application that provides user with football information such as league table, statics, player information like age, height, position and most importantly player valuation using to model that I created XGboost model and Linear and random forest tree model.

3. Methodology

This section outlines the complete development process of the project, from data sourcing to deployment. It includes the solution methodology, system requirements, and the software tools used. The methodology details how datasets were combined, how the model was built and evaluated, and how it was deployed as part of an interactive web application. It also defines the system's functional and non-functional requirements, and lists the technologies used across the frontend, backend, and machine learning pipeline.

3.1 solution Methodology

Model developed using datasets from Transfermarkt and Football Manager. Using this dataset will provide important data about the player's abilities, market data, and real-life statistics, helping to make a model that can predict players' valuations accurately.

3.1.1 Datasets

1. **Football Manager Data**: This dataset includes player characteristics, skill ratings, and potential, all rated through the lens of an in-depth video game simulation. Football Manager data is often detailed, capturing attributes like passing, shooting, and positioning, which are valuable for assessing a player's potential and impact.

2. **TransferMarkt Data**: TransferMarkt provides real-world transfer values, historical fees, player ages, contract lengths, and club affiliations. Which will help add an extra layer to the relevance of the model.

3.1.2 Algorithms

Two models were created applying two separate algorithms:

Using the strengths of both Algorithms, this model will mix Linear Regression with Random Forest Regression to forecast football player transfer values.

- The baseline model is linear regression, which detects linear links between market value and player characteristics (e.g., age, goals, assists). It offers interpretability, so guiding knowledge of how specific components affect player valuation.
- Through several decision trees, Random Forest Regression captures non-linear relationships so improving the model. Through averaging predictions, it lowers overfitting and increases accuracy by considering intricate interactions between features.
- Extreme Gradient Boosting, XGBoost will be applied for its efficiency and capacity to manage vast, datasets. Particularly in a dataset including many features like Football Manager and Transfermarkt, XGBoost can provide high predictive performance and capture intricate trends.

3.1.3 The Workflow

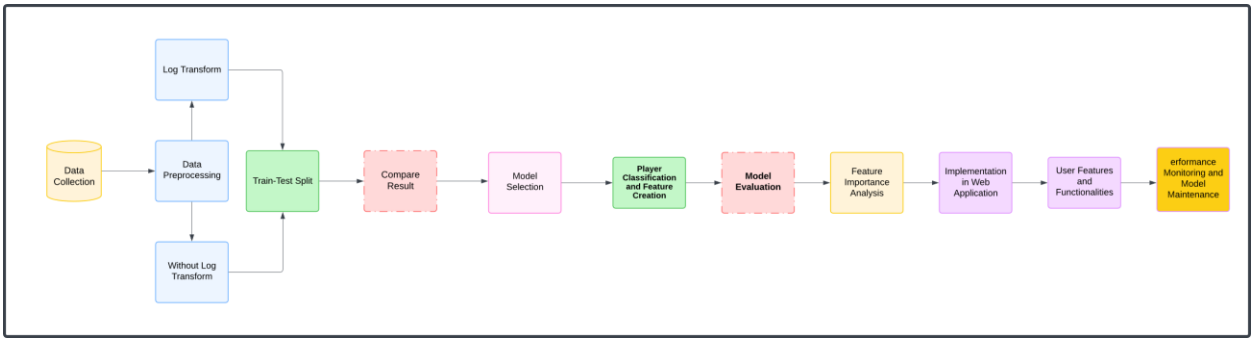


Figure 3 Workflow of the model

➢ **Data Collection**

- Data is collected from Football Manager and Transfermarkt. This data includes information about players' abilities, real-life data, and market data. creating a foundational dataset for building the model.

➢ **Data Preprocessing**

- Data is cleaned and organized. This step includes removing duplicate data, handling missing values, and preparing the dataset for the operation that is going to be implemented on it.

➢ **Log Transform / Without Log Transform**

- Two different paths are applied here: one with a log transformation and one without it. The log transformation is used to reduce skewness in data like transfer fees, making the distribution more uniform.

➢ **Train-Test Split**

- The datasets are divided into two sets training and testing. The model trains using one part of the data and tests on the other part to evaluate performance, this split is to make sure to prevent overfitting.

➢ **Compare Result**

- The results of the log-transformed data and the non-transformed data are compared to determine which form of data will be used going forward. The log-transformed proved better results

➢ **Algorithm Selection**

- Different algorithms will be tested to find the best one for predicting transfer fees. This step involves experimenting with various models and selecting the one that performs best on the data.

  The model will utilize multiple machine-learning algorithms, chosen for each one ability to handle different features:

  - **Model Evaluation**

  - The chosen model is evaluated using metrics like R^2, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and **mean absolute percentage error** (MAPE). To check the model accuracy in predicting market value of players

  - **Player Classification and Feature Creation**

  - Age was calculated by subtracting the birth year from 2025 and added as a new feature.

  - **Contract Years Remaining** was computed by comparing the contract expiration year with 2025.
  - A **classification function** was applied to group players based on multiple factors such as: Age group (Young, Experienced, Veteran), Playing position, Ability and potential ratings League level, Footedness, height, Contract status and Reputation level

  - **Feature Importance Analysis**

  - This step analyzes which player attributes (e.g., age, goals, assists) are the most influential in predicting transfer fees. Understanding feature importance provides insights into what drives player valuations.

  - **Implementation in Web Application**

  - A web application was designed and implemented using VUE.

  - **Implementation** of backend end Api
  - The backend will store the user's information in a MongoDB
  - Uses an Api Football-data.org Api that provides stats and match schedules
  - Uses a Transfermarkt Api that is locally hosted on docker to provide information such as current clubs, injures and trophies
  - **User Features and Functionalities**

  - New features will be added to the web app such as Customizable player Valuation. To help improve the user's experience.

➢ **Performance Monitoring and Model Maintenance**

- In this step the model's performance accuracy will be monitored, trying to improve the model.

3.2 Functional/ Non-functional Requirements

The Model will be accessible through a website build VUE that will provide the users with UI to interact with the model it will include multiple useful feature

3.2.1 Functional Requirements:

1. **Sign-In & Registration:** the user will be able to make an account to make account and sign in to it using their credentials
2. **User Profile:** Users will be able to edit their personal information
3. **Player Search:** The user will be able to use the search feature to search for a specific player
4. **Player & Team Profiles:** Each Team or player will have a specific profile that contains different information about the club or the player
5. **Customizable player Valuation**: Allows users to adjust the various attributes (e.g., age, position) to see how different it affects the player's market valuation. Using search algorithms that averages the top 3 player values to estimate the player price at his highest market value.
6. **Today's Matches:** Users will be able to view a list of all football matches scheduled for the current day, including details such as teams, match time, and competition
7. **Article Reading:** Users can browse and read football-related articles published on the website, including transfer news, match analysis, and player insights.
8. **Return on Investment (ROI) Calculator**: calculates whether the transfer fee paid for a player was worth it or not. compares the player's transfer fee paid to their Transfer market value.
9. **Reddit Word Cloud Generator:** The user enters the subreddit they want to search in and provides a keyword related to the topic of interest. The system generates a visually appealing word cloud highlighting the most frequently used words.
10. **Squad Builder:** The user selects a total budget from predefined options and then builds their ultimate football squad by selecting players for each position on a virtual football pitch. Players can be filtered and sorted by attributes such as age, predicted market value, league, or name. The system prevents duplicate players in different positions and calculates the total squad value dynamically, alerting the user if the selected squad exceeds the available budget.

3.2.2 Non-Functional Requirements:

1. **Performance**: The Web application should be able to search for a specific user without latency.
2. **Scalability**: The system will be designed to withhold increased user demand.

3. **Usability**: The UI will be user-friendly, designed using VUE, to ensure ease of navigation for all the users.

3.3 Software

Here is the list of all the software used and why it was used

3.3.1 Python: Because of its extensive libraries for data science, simplicity of use, and flexibility in experimentation, Python was the main programming language used in building the machine learning pipeline.
3.3.2 Pandas: Pandas was applied for effective transformation and data manipulation. It was absolutely important for handling missing values, cleaning the data, and creating fresh features.
3.3.3 NumPy: Particularly during feature scaling and mathematical modeling, NumPy was used for vectorized computations and handling of numerical operations.
3.3.4 Scikit-learn: supplied a broad spectrum of supervised and unsupervised machine learning techniques. It was applied in cross-valuation, and training of regression models.
3.3.5 XGBoost / LightGBM: Using their outstanding performance in managing structured tabular data and generating high-accuracy market value predictions, XGBoost and LightGBM were boosted algorithms were applied.
3.3.6 Matplotlib and Seaborn: Visualizing data distribution, correlations, and model outputs including feature importance and error residuals made use of Matplotlib and Seaborn.
3.3.7 Flask: The trained machine learning model's lightweight REST API was created using Flask.
3.3.8 Docker: The Transfermarkt clone API as well as the Player Similarity API were housed in Docker. This guaranteed consistent deployment over environments and streamlined backend service hosting.
3.3.9 MongoDB: User information, player profiles, market values, user-created squads, and saved comparisons effectively and in a scalable manner using MongoDB—a NoSQL document database.

3.3.10 Node.js: Node.js was used to manage server-side logic for user authentication, API routing, and communication with MongoDB. It worked as a middle layer between the frontend and database.

3.3.11 Express.js: Express.js, a minimalist web framework for Node.js, was used to structure and handle API endpoints for data retrieval, storage, and user interaction.

3.3.12 Vue.js: Vue.js was used to build the frontend interface of the Transfermarkt-style web application. It allowed dynamic rendering of player cards, search filters, squad builder UI, and API integration.

3.3.13 Axios: Axios was used in the Vue.js frontend to perform HTTP requests to the Flask and Node.js backends, allowing real-time data flow between the server and client.

3.3.14 Insomnia**: Insomnia** was used during the development and testing phase to manually test REST API endpoints and ensure correct request/response behavior.

3.3.16 Google Colab: Google Colab was used to train and validate the ML models in a cloud environment with GPU acceleration, removing the dependency on local computational resources.

3.3.17 railway: was used to host the backend API that handles the user and player information

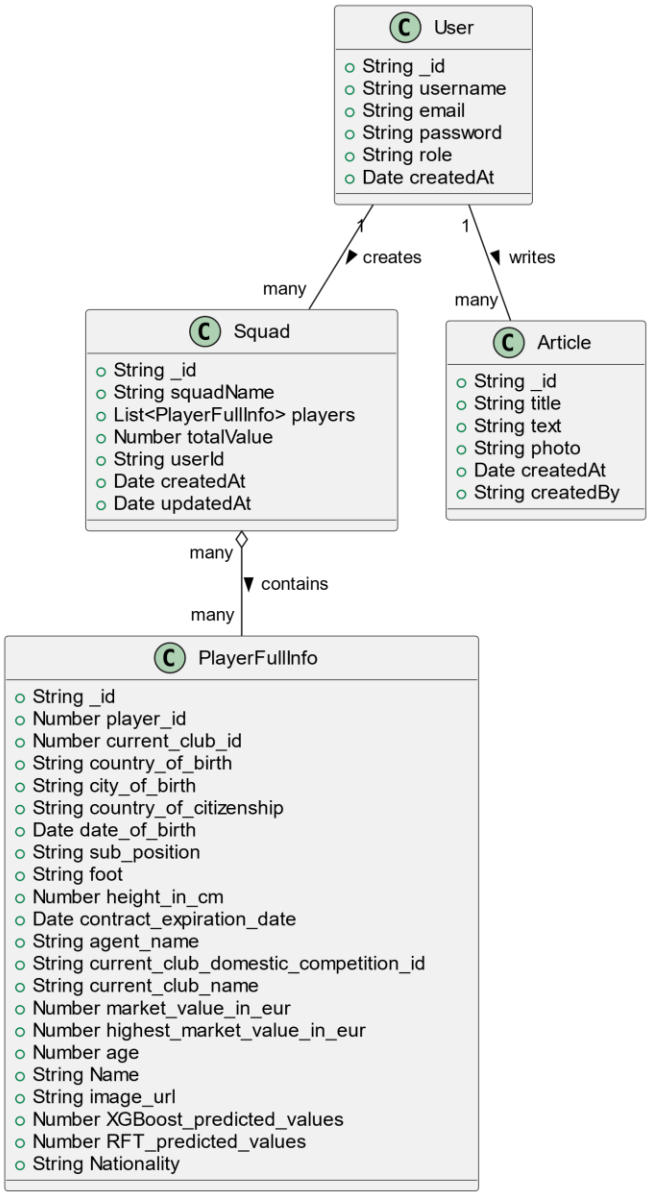
4. Diagrams & Design

This section provides visual representations of key processes within the system. The included flowcharts and UML sequence diagrams illustrate how users interact with the platform and how the system handles requests behind the scenes. Also includes screenshots from the web application.

4.1 Diagrams

This subsection will display all the UML diagrams that was made with explanation of the diagram

4.1.1 Class Diagram



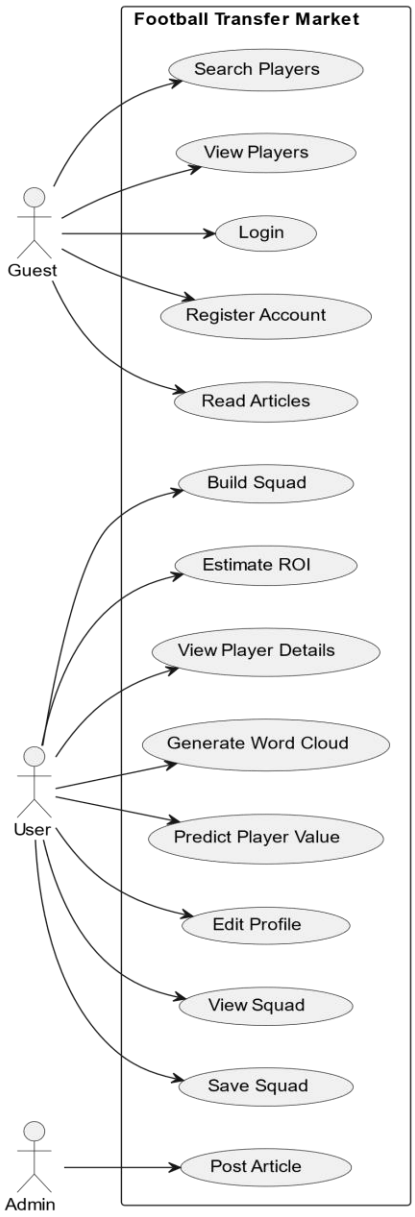
This class diagram illustrates the main data entities and relationships within the football transfer market platform.

- **User**: Represents registered users of the system. Each user can create multiple squads and write multiple articles.
- **Squad**: A user-generated selection of players. Each squad includes a list of players, a total squad value, and is linked to the user who created it.
- **PlayerFullInfo**: Contains detailed information about each football player, including personal details, club history, market value, and predicted values from different models. Players can exist independently and may or may not be part of a squad.
- **Article**: Represents articles written by users. Each article includes a title, content, optional photo, and metadata such as the creation date and author.

The relationships show that:

- A **User** can have many **Squads** and **Articles**.
- A **Squad** can contain many **PlayerFullInfo** entries.
- **PlayerFullInfo** exists as an independent entity and is not dependent on other classes.

4.1.2 System Use Case Diagram

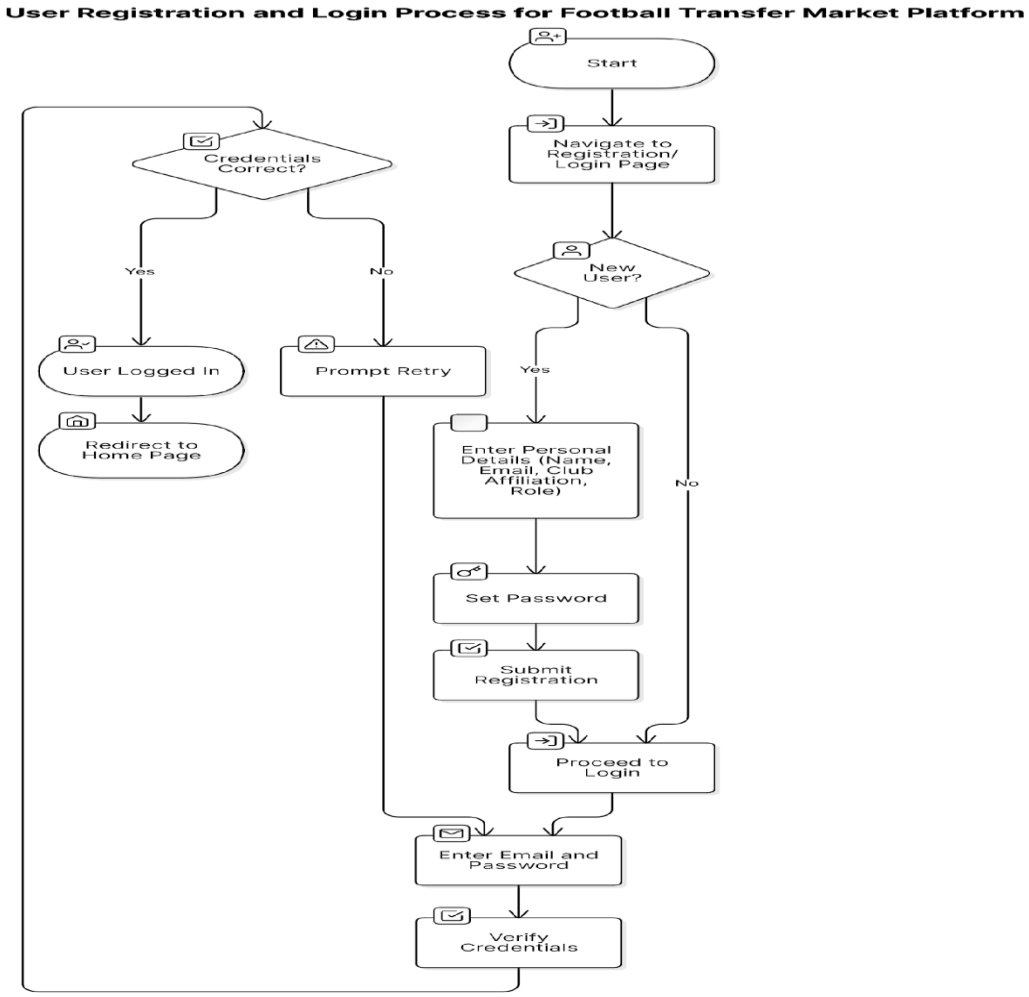


This use case diagram illustrates the core functionalities of the Football Transfer Market System and how different types of users interact with it.

- **Guests** can view general content such as player profiles, articles, and match information without logging in.
- **Registered Users** gain access to personalized features, including the ability to predict player market value, build and save squads, generate word clouds, and use the ROI calculator. They can also edit their profiles and explore deeper player analytics.

- **Admins** have additional privileges, such as creating and publishing articles for users to read.

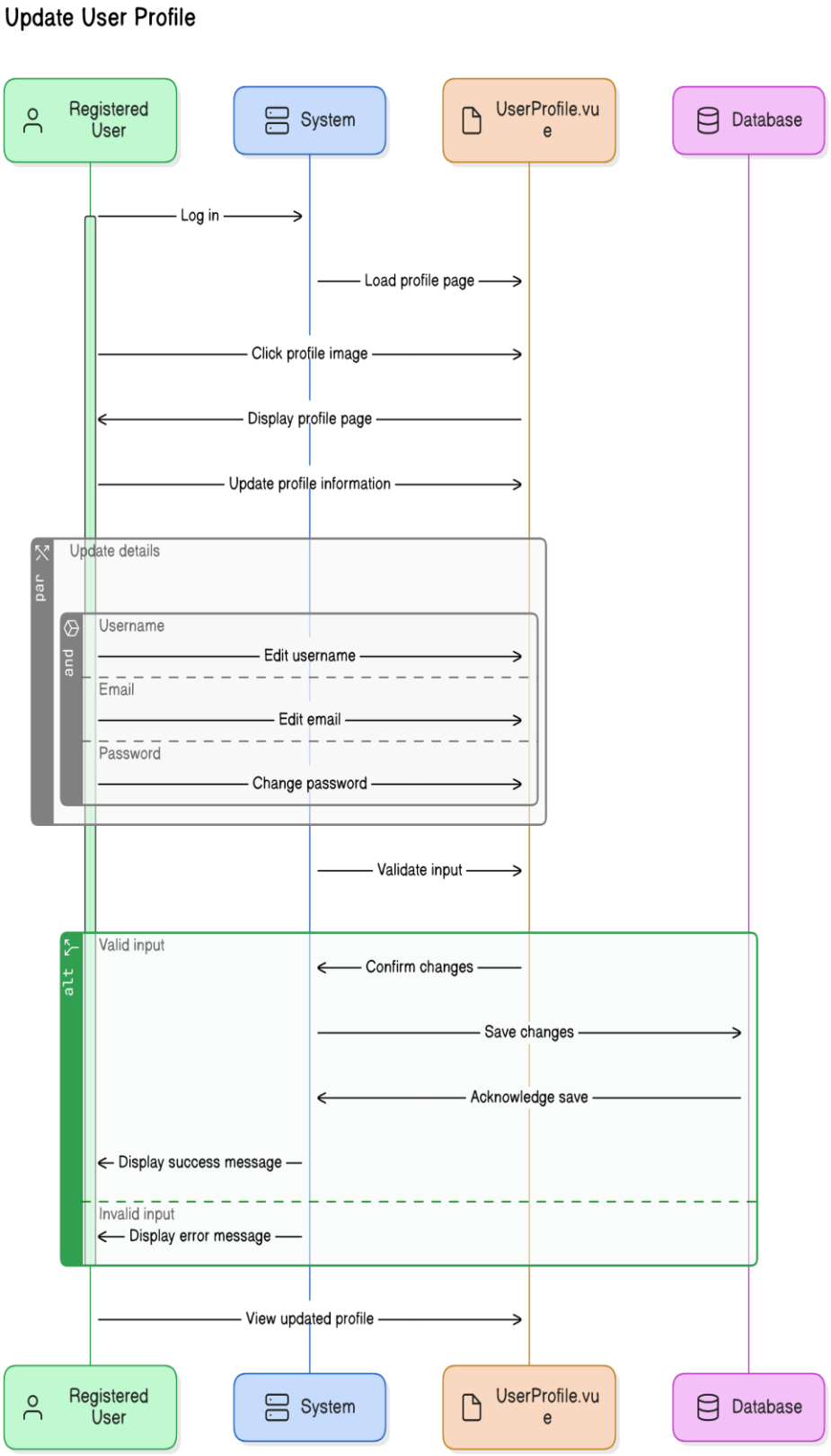
4.1.3 Flowchart of User Login/Register



This flowchart illustrates the process for both new and existing users interacting with the login and registration system.

- **Existing users** are directed to enter their credentials and, if correct, are granted access to the homepage.
- If credentials are invalid, users are prompted to retry.
- **New users** go through a registration process where they provide personal details, set a password, and then proceed to login.
- After successful verification, all users are redirected to the homepage.

4.1.4 Sequence Diagram of Update Profile

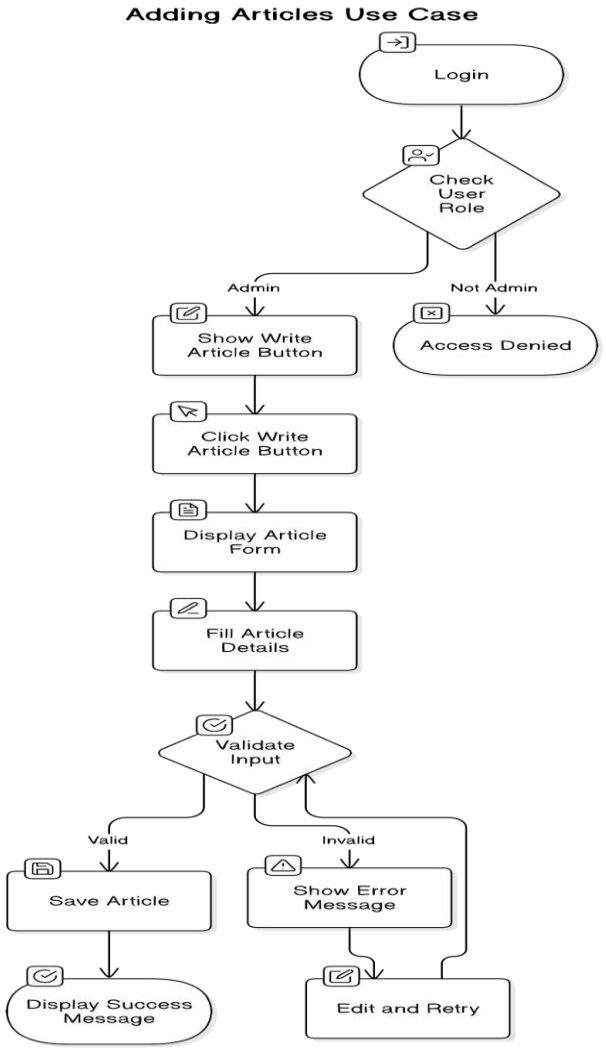


Step	Actor	Action	System Response
1	Registered User	Logs into the system	System loads user session
2	Registered User	Navigates to profile page	System loads UserProfile.vue view
3	Registered User	Clicks on profile image or update button	System displays current profile info
4	Registered User	Edits fields (username, email, password)	UI updates form inputs
5	System	Validates updated input	Checks for format, uniqueness, length, etc.
6	System	Confirms valid input	Saves changes to the database
7	Database	Receives update request	Updates the user record and acknowledges
8	System	Returns success message to UI	Informs user of successful update
9	Registered User	Sees confirmation	UI refreshes with new profile details

This sequence diagram shows how a registered user updates their profile on the platform.

- The user logs in and accesses their profile page via the frontend (UserProfile.vue).
- After editing fields like username, email, or password, the system validates the input.
- If the input is valid, changes are confirmed and saved to the database.
- A success message is displayed to the user. If there's an issue, an error message is shown instead.
- Finally, the updated profile is displayed, reflecting the saved changes.

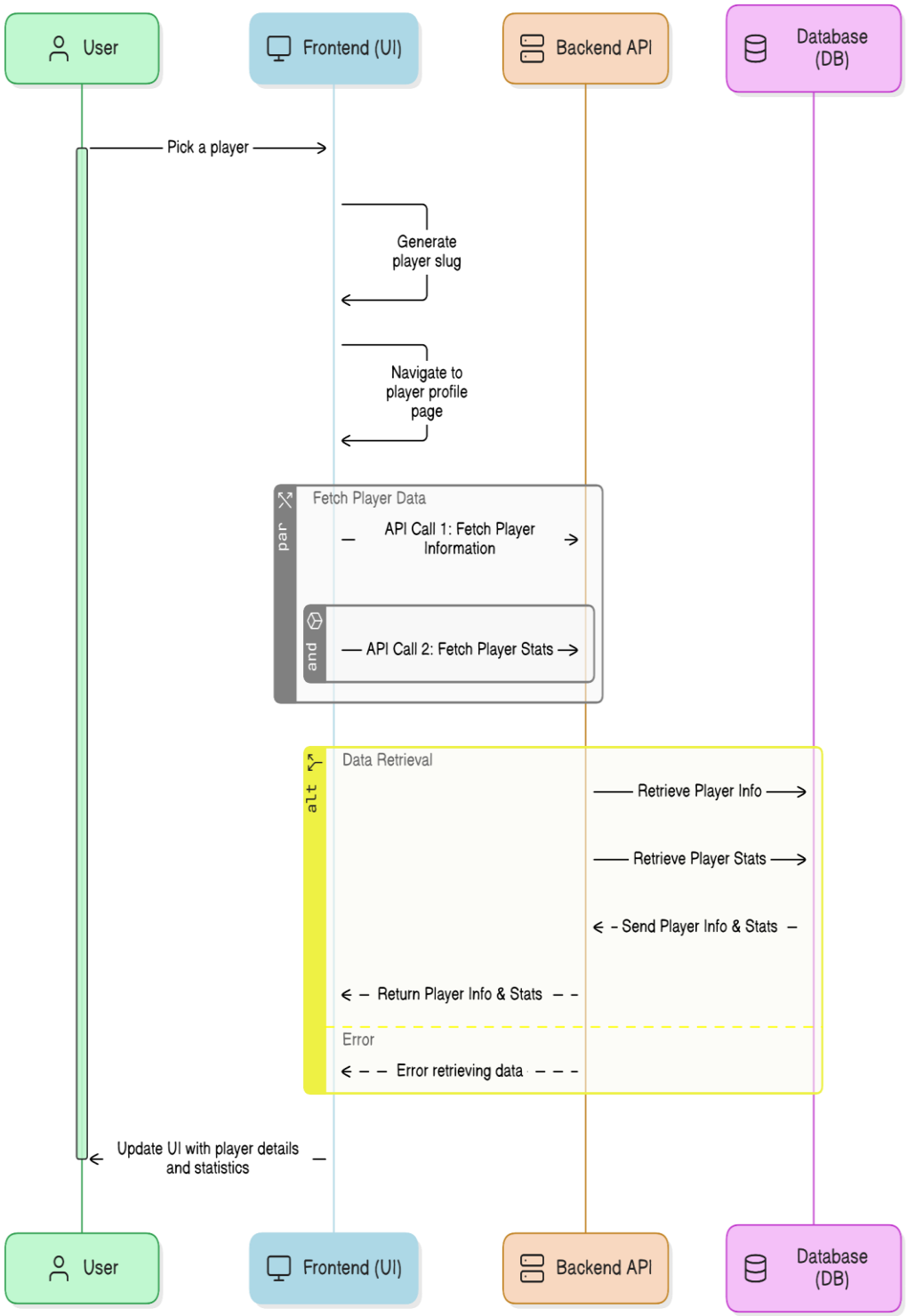
4.1.5 Flowchart of adding **article**



This flowchart outlines the process an admin follows to publish a new article on the platform.

- After logging in, the system checks the user's role.
- Only users with the **admin role** are shown the "Write Article" button; others are denied access.
- The admin clicks the button to open the article form, where they fill in details like title, text, and optionally a photo.
- The input is validated:
    - If valid, the article is saved and a success message is shown.
    - If invalid, an error message prompts the admin to correct the input and try again.

4.1.6 Sequence Diagram of Fetching Player Profile

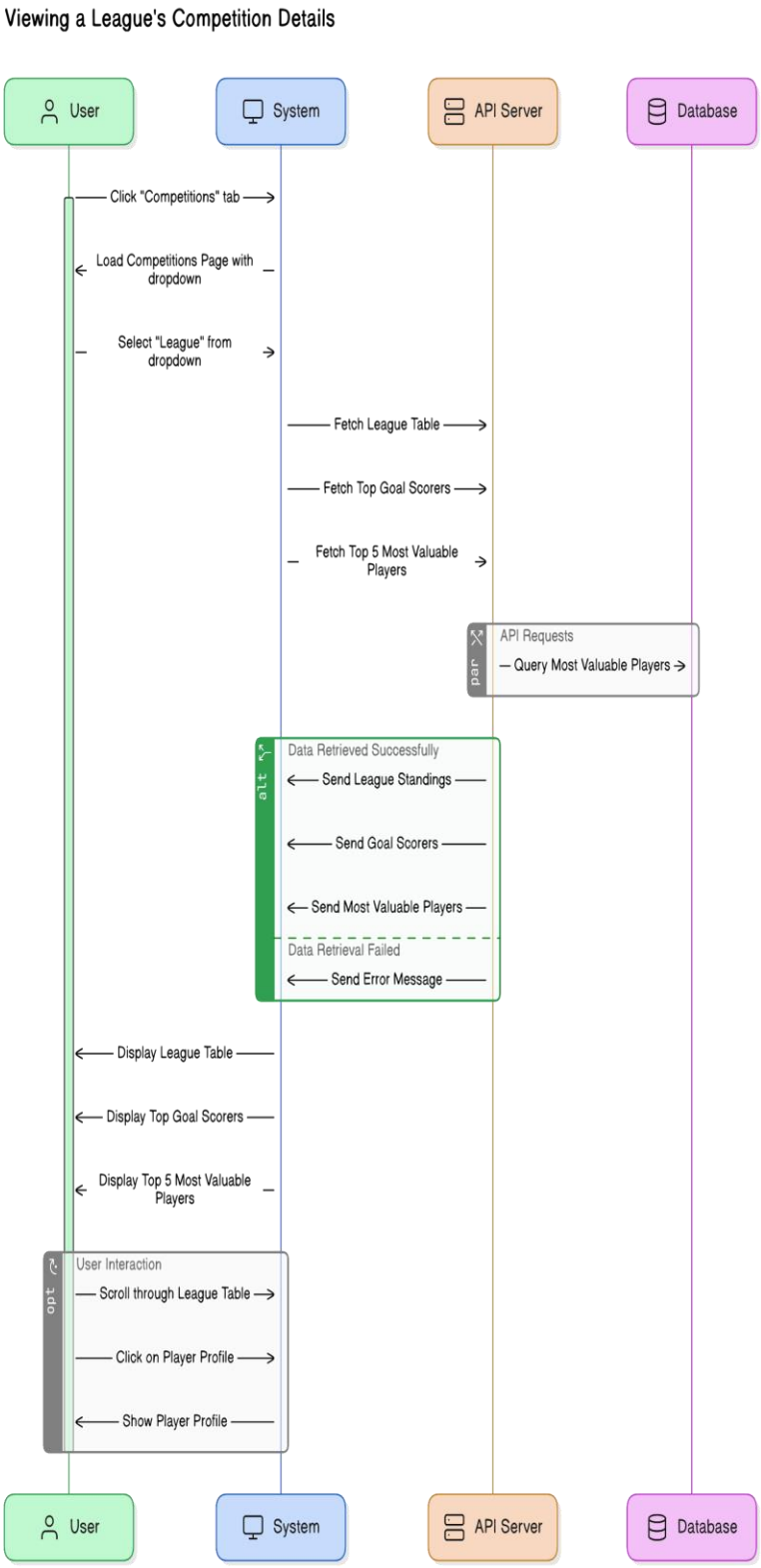


Step	Actor	Action	System Response
1	User	Picks a player	Triggers UI to generate the player's unique slug
2	Frontend (UI)	Navigates to the player profile page	Initiates API calls to retrieve player info and stats
3	Frontend (UI)	Sends API Call 1: Fetch Player Info	Backend receives request
4	Frontend (UI)	Sends API Call 2: Fetch Player Stats	Backend receives request
5	Backend API	Calls Transfermarkt API to scrape player info	Transfermarkt API scrapes and returns player bio and contract
6	Backend API	Calls Transfermarkt API to scrape player stats	Transfermarkt API scrapes and returns performance data
7	Frontend (UI)	Receives player info and stats	Updates Front-end with player details and statistics
8	*Alternate Path*	Error fetching from Transfermarkt or timeout	Displays error message or fallback Front-end

This sequence diagram shows how a user accesses a player's profile and how the system fetches the necessary data.

- The user selects a player, and the frontend generates a URL slug for the profile.
- The system navigates to the player profile view and initiates two API calls:
    - One to fetch general player information (e.g., name, club)
    - Another to fetch player statistics (e.g., goals, appearances)
- The backend retrieves the data from the database and sends it back to the frontend.
- If the retrieval is successful, the UI updates with the player's stats and details.
- If an error occurs during data fetching, an error message is shown.

4.1.7 Sequence Diagram of Fetching competition



Step	Actor	Action	System Response
1	User	Clicks "Competitions" tab	System loads the Competitions page with league selection dropdown
2	User	Selects a specific league from dropdown	System initiates data fetch requests
3	Front-end	Sends request to API: Fetch League Table	API Server queries the database
4	Front-end	Sends request to API: Fetch Top Goal Scorers	API Server queries the database
5	Front-end	Sends request to API: Fetch Top 5 Most Valuable Players	API Server scrapes/queries external source or DB
6	Backend API	Retrieves league standings	Sends league standings data to the system
7	Backend API	Retrieves top scorers	Sends goal scorer data
8	Backend API	Retrieves top 5 most valuable players	Sends most valuable players data
9	Front-end	Displays league table	Updates UI with current standings
10	Front-end	Displays top goal scorers	Shows list of top scorers in the league
11	Front-end	Displays most valuable players	Renders top 5 most valuable players
12	*Alternate*	API fails to retrieve data	Displays error message

This **UML Sequence Diagram** shows the user selecting a league, which triggers API calls to fetch the **league standings, top goal scorers, and most valuable players**. The system then displays the data to the user.

4.2 Design

This part will include screenshot from the web application and final result from the design process. And will list all the features inside the page.

4.2.1 Home Page

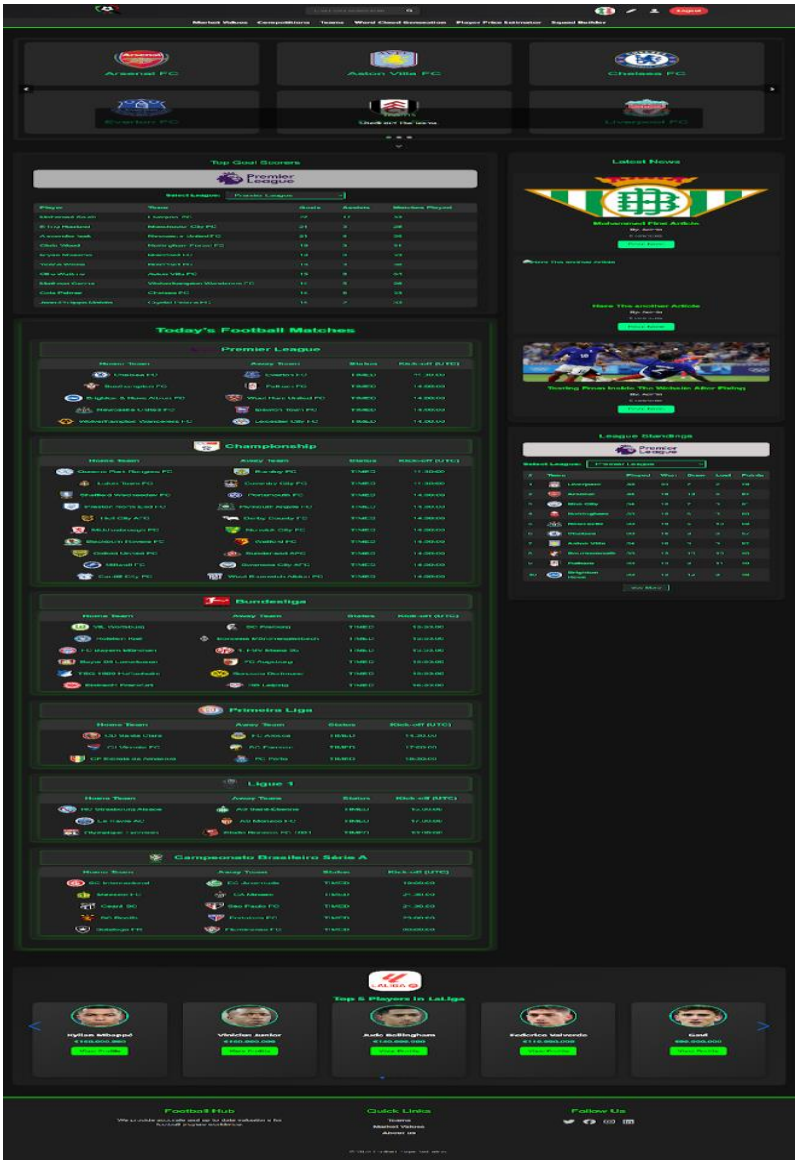


Figure 9 Home Page

On this page, the user can:

- View the top goal scorers across the top 5 leagues
- See the league table standings for each of the top 5 leagues
- Explore the top 5 most expensive players in each league
- Click on the **Articles** section to read football-related content
- Check the list of matches scheduled for today

4.2.2 Player Page

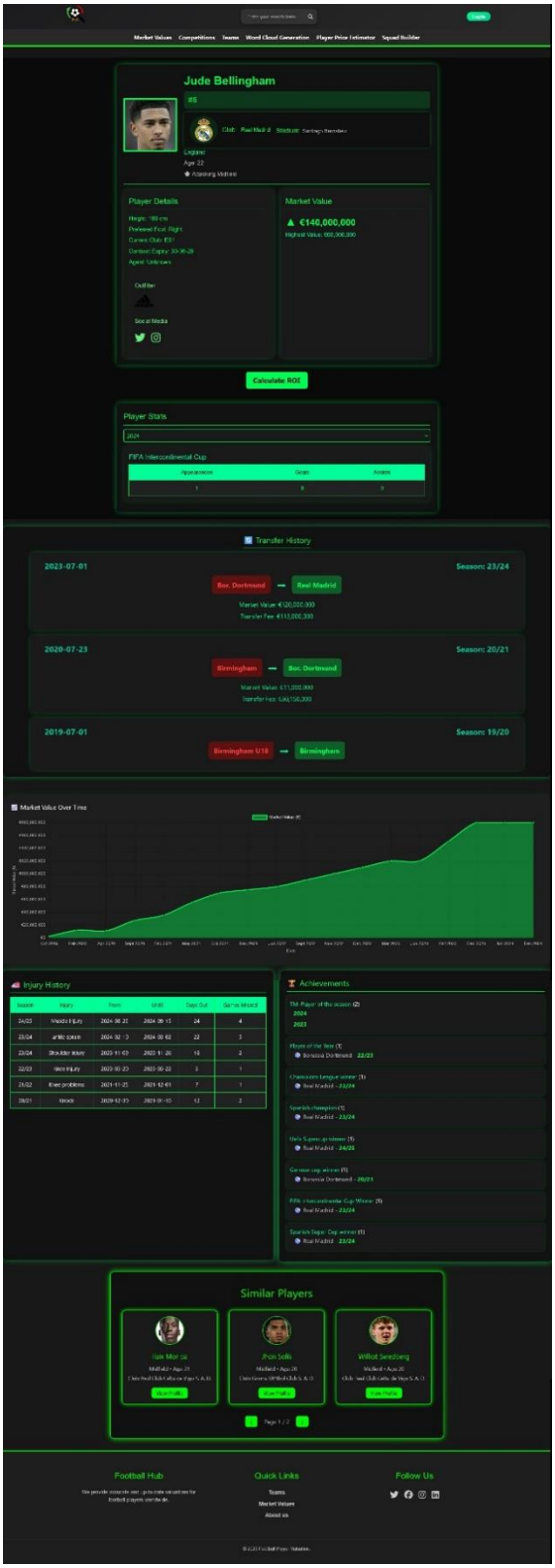


Figure 10 Player Page

On this page, the user can:

- View the player's general information, including:
    - Current club and home stadium
    - Outfitter brand (e.g., Nike, Adidas)
    - Agent name
    - Preferred foot
    - Social media links
- See player market value estimates from (via pop-up):
    - XGboost model
    - Linear & Random Forest model
    - Transfermarkt (real-world reference)
- Explore detailed stats such as:
    - Number of goals and assists
    - Trophies won
    - Injury history
- Analyze a **market value progression graph**, showing how the player's value has changed over the years
- check the **player's transfer history** including past clubs and fees involved.
- Check Similar Player from the same league how have similar attribuites.

4.2.3 Article view as admin



Figure 11 Article View

This is the article view page from the admin's perspective. In addition to reading the article content, the admin is provided with two key functionalities: the ability to **edit** or **delete** the article using the respective buttons below the content. This allows for quick management of published articles directly from the interface, supporting real-time updates or content removal when needed. Also the name of the author is shown as well the date of the article.

4.2.4 Team View

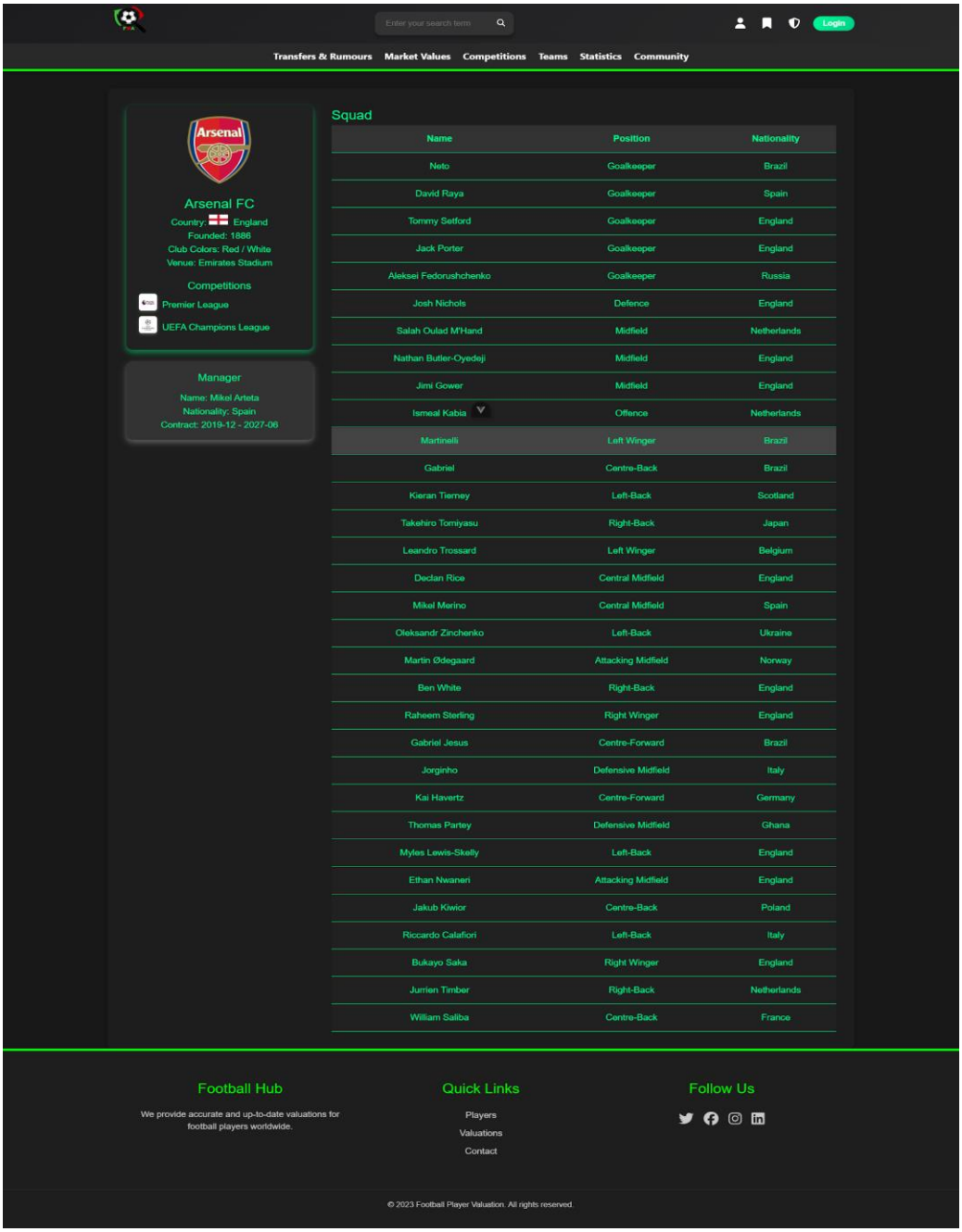


Figure 12 Team View

This page displays the full squad of a football team, listing each player's name, position, and nationality. On the left side, it also provides general information about the club such as the country, year founded, stadium, and the manager's name and contract details. Additionally, it highlights the competitions the team participates in—like the Premier League and UEFA Champions League—giving users a complete overview of the club's profile and lineup.

4.2.5 Squad Builder

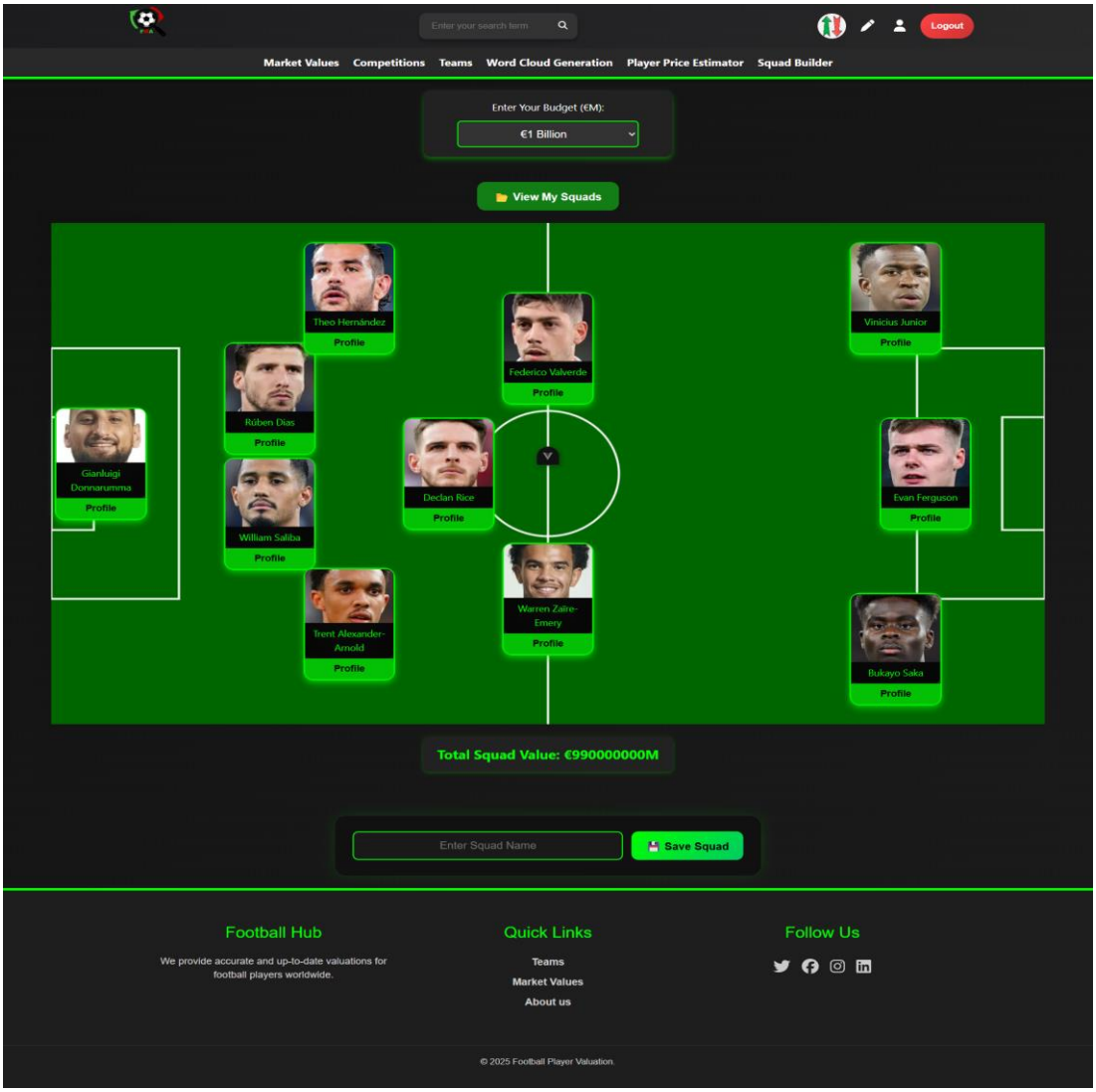


Figure 13 Squad Builder

This Squad Builder page allows users to make a custom football starting eleven by allowing the user to select players for each position through interactive pop-ups. Users can filter players based on specific criteria to find the right fit. The total squad value is calculated in real time and displayed at the bottom. Users can also set a budget limit if the total squad value exceeds the budget, a warning message appears. Once the squad is complete, users can assign it a custom name and save it for later.

4.2.6 Player value Estimator



Figure 14 Player Value Estimator

This page allows users to estimate a football player's market value by entering key attributes such as age, position, sub-position, preferred foot, height, league, and remaining contract years. After clicking "Predict Value," the Api that hosts the KNN model returns the predicted market Additionally, it identifies and shows three real-world players with the most similar profiles, including their market values and similarity percentages

4.2.7 Word cloud Generator from Reddit

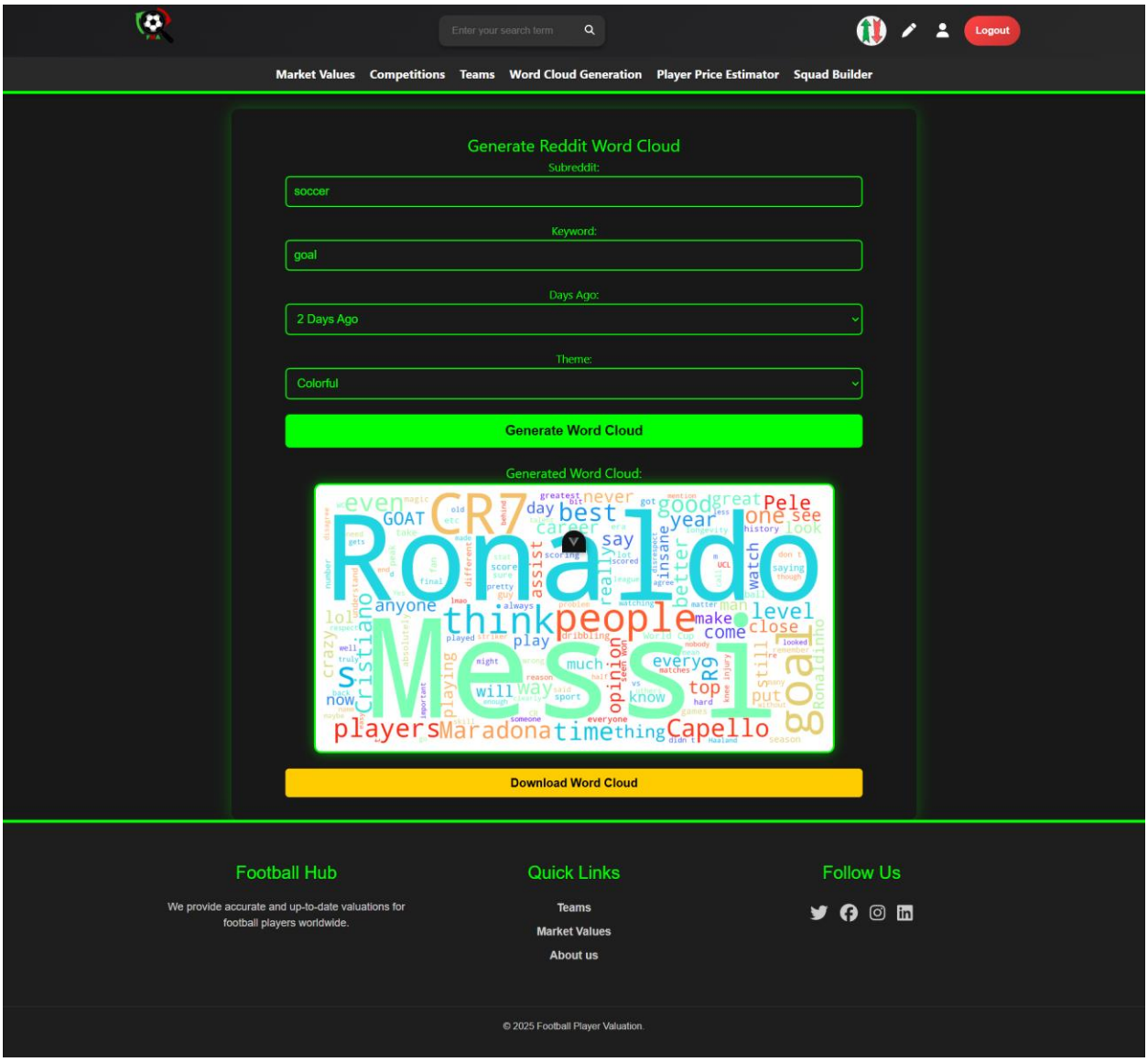


Figure 15 Word Cloud View

This page allows users to generate a word cloud from Reddit posts by entering a subreddit and a keyword. After selecting a time frame and a theme, users can click "Generate Word Cloud" to visualize the most frequently used words related to the topic. The generated word cloud highlights key terms discussed in the selected subreddit, helping users quickly grasp trending discussions. A "Download Word Cloud" button is also available, enabling users to save the visualization locally.

5. Implementation

This section details the step-by-step implementation of the football player market value prediction project.

5.1 Dataset Collection

Finding the required datasets is the first step in the project. From Kaggle, two important datasets were picked both providing insights on player performance and value:

In-depth simulated player data including skill ratings, physical and mental attributes, potential, and positions makes up Football Manager 2023 Dataset (FM2023). Often used in scouting simulations, it offers a detailed perspective of a player's attributes.

Real-world data including current player market values, past transfer fees, age, contract expiration dates, injury history, and club affiliations makes up the Transfermarkt Dataset (TM). For estimate of market value, it provides the ground truth.

Apart from the Kaggle datasets, I also included information to my dataset using a locally hosted Transfermarkt API housed on Docker. Not directly available in the Kaggle dataset, this API supplied dynamic and historical data. Using a Python script, the following further qualities were gathered and combined into the final dataset:
Useful for market value adjustment, player injury history includes length and recurrence of injuries.
Player Historic Transfer Values: Time-series information of a player's valuation development.

The datasets contain 2989 rows and consist of the following columns player_id current_club_id country_of_birth city_of_birth country_of_citizenship date_of_birth sub_position foot height_in_cm contract_expiration_date agent_name current_club_domestic_competition_id current_club_name market_value_in_eur highest_market_value_in_eur age Name image_url XGBoost_predicted_values RFT_predicted_values Nationality.

5.2 Dataset Merging Process

To be able to merge both datasets **a Python script using fuzzy name matching is used.** This script enabled more accurate alignment of player records across datasets, even in cases of slight spelling differences or abbreviations. The fuzzy matching algorithm evaluated similarity scores and linked players who met a defined confidence threshold.

5.3 Result of the Merge

One single csv file contains all the data such as player attributes, historical player value over the year and statistics that is needed to start implementing the model

5.4 Data Preprocessing

Once the datasets were successfully merged, a thorough **data preprocessing** phase was carried out to prepare the data for machine learning modeling. This step ensured the dataset was clean, consistent, and structured appropriately for training algorithms.

5.4.1 Data Cleaning

During this stage, duplicate entries, irrelevant columns, and inconsistent data were removed or standardized.

5.4.2 Handling Missing Values

Due to the integration of multiple sources (Transfermarkt, Football Manager, and the Transfermarkt API), some missing values were inevitable. These were handled using different strategies based on the type and importance of the data:

- **Numerical features** (e.g., transfer fee, appearances):
  Filled using statistical methods such as mean or median imputation depending on the feature distribution.

- **Categorical features** (e.g., position, nationality):
  Filled with the most frequent category or marked as "Unknown" when necessary.

5.5 Model evolution and Implementation: Linear Regression and Random Forest Ensemble

In this section the model evolution and how it was implemented will be explained. To begin model development, an ensemble approach was implemented combining **Linear Regression**, **Random Forest**, and **Gradient Boosting Regressors** to predict football players' market value. This model used feature-engineered data derived from the merged dataset.

5.5.1 Feature Selection and Engineering

To really capture what influences a football player's market value, I went beyond just using raw stats. I transformed values like transfer fees and market worth using log scaling so that outliers wouldn't skew the results. I also looked at how different features interact—for example, how a player's age combined with their ability might say more than either one alone. I included key performance ratios like goals per appearance and assists per goal to reflect on-field impact. To cover the bigger picture, I added features for injuries and transfer history, like how often a player gets injured or changes clubs. I even considered whether their value has been rising or falling recently. To make sure everything worked well together, I encoded positions into numbers and scaled the rest of the data so no single factor dominated. All of this helped create a more balanced and insightful model.

5.5.2 Model Training and Evaluation

The dataset was split into **80% training** and **20% testing**. Three models were trained independently:

- **Random Forest Regressor** with hyperparameter tuning via RandomizedSearchCV

- **Gradient Boosting Regressor**

- **Linear Regression** model for baseline comparison

Then, using optimal weights that reduced the Mean Absolute Error (MAE), an optimization function blended the forecasts from all three models. This combined approach captured both linear and non-linear connections, so improving general performance.

5.5.3 Model Design Evolution and Comparison

The ensemble modeling strategy was developed in two distinct stages.

In the early version, three base models were trained:

- A **Linear Regression model** to provide interpretability and establish a baseline
- A **Random Forest Regressor** to capture non-linear interactions using bootstrapped trees
- A **Gradient Boosting Regressor** to refine the decision boundaries using sequential boosting

These models were blended using equal weights to create an initial ensemble. However, this model was limited by its reliance on a basic feature set and lacked more in-depth features like injuries or recent performance.

5.5.3.1 Comparison between Evaluation metrics of the both model

In this section a small comparison between the old version of the model with the Final model.

Evaluation unit	Old Model	Final improved model
Ensemble MAE	3,317,876.17	3,095,660.94
Ensemble RMSE	—	6,732,595.77
Ensemble R-squared	0.827	0.878
Ensemble MAPE	—	48.92 %

5.6 Model Evolution and Implementation: XGBoost Algorithm

In this section the model evolution and how it was implemented will be explained.

5.6.1 Feature Engineering and Enhancements

This implementation began by engineering additional performance, contract, and league-related features:

- **Contract- and league-interaction features**, such as contract_x_league, ca_x_league_strength, and league_pa_interaction
- **Discipline and contribution metrics**, such as weighted_yellow_cards, goal_to_appearances_ratio, and assist_to_appearance_ratio

- **Market value trend indicators**, such as last_6_month_change, last_1_year_change, and log_highest_market_value_normalized
- **Injury impact features**, quantifying severity through injury_impact_score and interaction terms like injury_x_age
- **Transfer dynamics**, including transfer_activity_score and transfer_x_league_strength

5.6.2 Model Training and Ensemble Strategy

Training and testing set totaling 80/20 comprised the dataset. Training the XGBoost Regressor with a hyperparameter tuning approach based on RandomizedSearchCV.

An ensemble weighting function was used to minimize the **Mean Absolute Error (MAE)** by finding optimal weights for combining predictions. This approach allowed the model to leverage the strengths of all three algorithms.

5.6.3 Model Evolution Summary

The XGBoost model underwent a two-stage evolution: from a basic model with core football metrics to an advanced system capturing detailed, real-world dynamics.

Initial Model – Basic Feature Set:

- Technical ratings such as **current ability (ca)** and **potential (pa)**
- Simple interaction terms like age_x_ca
- Basic indicators including transfer fee history, reputation scores, and positional encoding

While getting good evaluation, this version needed some extra features to improve the model ability to predict even more.

Improved Model – Advanced Feature Set:

- Context-aware features such as contract_x_league and league_pa_interaction
- Ratios reflecting individual match contributions (e.g., goal and assist metrics)
- Red/yellow card weightings by position to quantify on-pitch discipline
- Injury and transfer interaction terms to model reliability and market movement volatility

5.6.3.1 Comparison between Evaluation metrics of the both model

Evaluation unit	Old Model	Final improved model
Ensemble MAE	3384210.423798524	2614963.723609162
Ensemble RMSE	7616636.898415847	5978582.450982708

Ensemble R-squared	0.8563586419461541	0.9050188764538948
Ensemble MAPE	54.31%	41.47%

5.7 Merging both models' outcome in one dataset

To further improve the quality and contextual accuracy of the dataset, a final enrichment step was applied to include **player nationality** and **age**, ensuring a more complete profile for each player in the dataset.

5.7.1 Merging Process of player market value estimation in one dataset

Two datasets were used in this step:

- Main Player Dataset (final_dataset.csv): This includes all cleaned and feature-engineered need for the model.
- Nationality Dataset (merged_data.csv): Contains player_id and corresponding Nationality fields.

The merge operation was performed using a left join on player_id, ensuring that every player in the main dataset retained their record, while nationality was added where available.

5.7.2 Age Calculation

If the column date_of_birth exists the age of each player were calculated using python datetime module by of the birth year from the current year.

5.7.3 Export and Output

The updated dataset was saved to Output file named final_dataset_with_nationality_and_age.csv

5.8 Backend API Development using Node.js and MongoDB

To support the web application and connect the machine learning model to user-facing features, a **RESTful API** was built using **Node.js** and **Express.js**. The API serves as the backend layer that handles authentication, user data management, player information storage, squad building, and more.

5.8.1 Backend Structure and Routing

The Express server is modularized using multiple route files to maintain clean architecture and scalability. Key routes include:

- /api/user – Handles user registration and login
- /api/players – Stores and retrieves predicted player information
- /api/search – Allows searching for players
- /api/fetchuser – Retrieves logged-in user data
- /api/squads – Manages user-created squads

- /api/articles, /api/football, /api/table – For news, match stats, and tables
- /api/handlePlayer – CRUD operations for player data
- /api/players/search-sort – Supports searching and sorting via middleware

Middleware functions such as searchMiddleware and sortingMiddleware were added to separate concerns for filtering and organizing player data.

5.8.2 Connecting to MongoDB

The backend connects to a **MongoDB** database using **Mongoose**, allowing storage and retrieval of structured documents such as:

- **User accounts** (email, password, profile info)
- **Predicted player values**
- **Saved squads**
- **Custom player uploads or edits**
- **Saved Articles**

5.8.3 Use Case Example

- When a user registers or builds a squad, the frontend sends a POST request to the respective route (e.g., /api/squads). The backend validates, stores, and retrieves the data from MongoDB, enabling persistent user interaction and saving predictions generated by the ML model.
- User registers to make an account

5.9 MongoDB scheme Designs

This section outlines the **MongoDB** schema designs used in the **backend API**. Each schema corresponds to a collection in the database and defines how different entities like users, players, and squads are structured, stored, and retrieved through the API.

5.9.1 User Schema Design

The User schema defines how user accounts are stored in MongoDB. It includes the following fields:

- username: A unique, required string (min 3 characters)
- email: A required, unique string validated as a proper email format
- password: A required string (min 6 characters), hashed before saving
- role: A string with values 'user' or 'admin' (default: 'user')

- profileImage: Optional string (image URL or base64)
- createdAt: Automatically stores the creation date

5.9.2 Player Schema Design

The Player schema stores full information about each football player, including both real-world data and predicted values from the machine learning models. It is used to manage player profiles, enable search and filtering, and support squad building.

Key Fields:

- player_id: Unique auto-incremented ID for each player
- Name, age, Nationality: Basic identity and demographics
- sub_position, foot, height_in_cm: Player role and physical attributes
- date_of_birth, city_of_birth, country_of_birth: Biographical data
- current_club_id, current_club_name, current_club_domestic_competition_id: Club details
- market_value_in_eur, highest_market_value_in_eur: Real transfer market values
- XGBoost_predicted_values, RFT_predicted_values: Predicted player value by each model
- image_url: Link to player's image
- contract_expiration_date, agent_name: Contract and agent information

The schema uses mongoose-sequence to automatically assign and increment player_id. It is connected to the PlayerFullInfo collection and timestamped for tracking creation and updates.

5.9.3 Squad Schema Design

The Squad schema stores user-created squads, allowing users to build teams and track their total value.

Fields:

- squadName: The name of the custom squad
- players: An array of player objects or IDs
- totalValue: The combined market value of the squad
- userId: Reference to the user who created the squad

This schema supports the **squad builder feature** in the frontend.

5.9.4 Article Schema Design

The Article schema allows admins or users to post football-related content such as transfer news or analysis.

Fields:

- title: Title of the article (min 5 characters)
- text: Main content body (min 10 characters)
- photo: Optional image related to the article
- createdBy: Reference to the user who wrote the article
- createdAt: Automatically set to the article's creation time

This supports the **Article Reading** feature, and adding new article if the user is an admin.

5.10 Uploading Final Dataset to MongoDB

After generating predictions using both the **XGBoost** and **Random Forest + Linear Regression** models, the final results were exported into a single CSV file. This dataset included all player information along with their respective predicted values.

To integrate this data with the backend API:

- The CSV file was **converted to JSON format**
- It was then **imported into a new MongoDB collection** using tools like MongoDB Compass.
- The new collection became part of the existing database structure and was made accessible through the API's /api/players route

This allowed the frontend and other services to retrieve predicted player data dynamically from the database, enabling features like player search, squad building, and valuation display

5.11 Hosting the API on Railway

My backend API was deployed using Railway, Railway simplifies deployment by integrating with GitHub and automating the build and deployment process.

The project's environment variables such as API keys were configured through Railway's dashboard. Once deployed, the API was publicly available and could be accessed by the Vue.js frontend.

5.12 Web Application using Vue

The Front-end of the web application was created using Vue.js. allows each page in the application represents a specific functional view, mapped to a corresponding component, and is organized using **Vue Router** for navigation.

5.12.1 Pages in the Application

Below is a list of the main pages implemented in the web application:

- **Home Page (HomeView.vue)**
  The main landing page of the application, showcasing key features or navigation links.

- **Signup Page (SignupView.vue)**
  Allows new users to register an account.

- **Login Page (LoginView.vue)**
  Enables existing users to log into the system securely.

- **About Us Page (AboutUs.vue)**
  Provides information about the application or the development team.

- **Player Profile Page (PlayerView.vue)**
  Displays detailed information about an individual player, including stats, bio, and predicted values.

- **Search Results Page (SearchResult.vue)**
  Shows players matching a search query or filter criteria.

- **Market Values Page (MarketValues.vue)**
  Presents a visual overview of predicted market values from the models.

- **Create Player Page (CreatePlayer.vue)**: Lets users manually input and submit a new player into the system.

- **Player Value Generator Page (PlayerValueGen.vue)**: Allows users to enter custom attributes and receive a predicted market value using ML models.

- **Word Cloud Page (WordCloud.vue)**: Generates a word cloud based on Reddit data for sentiment analysis or trending topics.

- **Squad Builder Page (SquadBuilderView.vue)**: Lets users build and customize football squads within a predefined budget.

- **My Squads Page (MySquads.vue)**: Displays saved squads created by the logged-in user.

- **Article Details Page (ArticleDetails.vue)**: Shows full content of a football-related article or blog post.

- **Today's Matches Page (TodayMatches.vue)**: Lists real-time matches scheduled for the current day.

- **Upcoming Matches Page (UpcomingMatches.vue)**: Displays upcoming matches for a specific competition or team.
- **Competition Page (Competition.vue)**: Provides standings, fixtures, or details about different football competitions.
- **Teams Page (Teams.vue)**: Lists available teams in the database with basic club info.
- **Team Profile Page (TeamProfile.vue)**: Displays information about a specific team, including its players and stats.
- **Edit Profile Page (EditProfile.vue)**: Allows users to update personal information such as name, email, or password.

5.13 Player Value Estimation via Similarity Matching (PlayerValueGen.vue)

A front-end dedicated Player Value Generator page was created to give user a more interactive experience. Users may enter player traits and get an estimated market value depending on the top 3 most similar players found in the dataset.

5.13.1 Algorithm Overview

The algorithm is implemented using FastAPI (Python) and leverages the K-Nearest Neighbors (KNN) approach to estimate market value based on player similarity. this system finds the most similar players and averages their real historical market values. This model is hosted locally on docker.

5.13.2 Key Steps

1. **Input Features**
   Users provide player-specific features such as: Age Position, sub-position, Preferred foot, Height, League and, Contract years remaining
2. **Data Filtering**
   The uploaded dataset (Final_Player_Data_With_Market_Value.csv) is filtered by: position, league and sub position
3. **Feature Normalization**
   Relevant numeric features (age, foot_encoded, height, contract years) are scaled using **MinMaxScaler** for uniformity in distance calculations.
4. **Nearest Neighbor Search**
   A **KNN model** (NearestNeighbors) identifies the top 3 most similar players based on the scaled feature space.
5. **Market Value Estimation**
   The average **highest historical market value** of the **3 similar players is calculated** and rounded to the nearest million euros. This is returned as the **estimated market value**.

6. **Response Output**
   The API returns both: estimated player value and the e most similar player to the user's input

5.13.3 Integration with the Frontend

The FastAPI endpoint /predict_market_value/ is accessed via the **PlayerValueGen.vue** page. Users fill a simple form, and upon submission, the Vue app sends a POST request to the backend and receives the predicted value and list of similar players.

5.14 Word Cloud generator

To enhance the visual exploration of fan sentiment, this project includes an API (hosted on docker) built with **FastAPI** that connects to Reddit via the PRAW (Python Reddit API Wrapper). The API generates **word clouds** from match-related Reddit threads, offering a word cloud of trending phrases associated with the user's inputed word, these visualizations can be filtered by subreddit, match keyword, date, and custom visual themes.

5.14.1 Key Steps in the API Workflow

1. **Reddit Connection** the API uses PRAW to securely connect to Reddit using credentials. It accesses a goes subreddit and searches in reddit posts based on a match-related keyword.
2. **Text Extraction** Once relevant threads are identified, the API gathers their titles, bodies, and the first few hundred comments. This content is cleaned from deleted posts, links, and non-informative noise.
3. **Words like** game, match, team is excluded from the search via stopwords list Storage and Cleanup Generated images are stored in a /wordclouds folder.
4. **Cleaning and Storage** The /wordclouds folder contains the generated images. To conserve storage, a cleanup feature automatically eliminates photos that are more than 24 hours old.

5.14.2 Integration with the Frontend

The frontend can make HTTP GET requests to the API's then the API responds with a direct **image URL**, which can be embedded in the frontend UI for real-time display.

This integration allows users to easily explore how fans are reacting to matches based on live Reddit discussions—offering insights into sentiment, hype, and emerging narratives.

5.15 Similar Players API (KNN-Based Recommendation)

This API serves player recommendations by identifying others with similar profiles based on players features.

5.15.1 Key Steps in the API Logic

Loads player data and generates years remaining on every player's contract.
Filtering: Restraints recommendations to players in the same level and position to keep relevance.
For equitable distance comparison, feature scaling normalizes age, preferred foot, height, and contract years.
Using K-Nearest Neighbors (KNN), find the top 5 most similar players (except from the current one).

5.15.2 Frontend Integration

The frontend makes a call to /similar_players/{player_id}. The API responds with a list of player profiles that are similar in style and attributes, making it easy to show similar player cards directly

in the playerView the API response includes name, club, position, age, and image URL for immediate display.

5.16 Transfermarkt API (Hosted Locally on Docker)

To retrieve more player-specific information beyond the static dataset, a custom **Transfermarkt API** was deployed locally using **Docker**. This API scrapes and serves detailed football player data directly from Transfermarkt's public profiles.

5.16.1 Purpose and Usage

The API is used to dynamically fetch up-to-date and historical player information, which improves the player profiles presented on the web application. This includes data not directly available in the main dataset.

5.16.2 Key Endpoints and Functions

The locally hosted API supports the following key methods:

- **fetchPlayerClubProfile**: Returns club history, contract info, and previous affiliations
- **fetchPlayerStats**: Retrieves player match statistics, including goals, appearances, and assists
- **fetchPlayerInjuries**: Provides a timeline of injuries, including type and duration
- **fetchPlayerJerseyNumber**: Returns the jersey number worn at the current club
- **fetchPlayerAchievements**: Lists individual and team achievements or trophies won
- **fetchTransferHistory**: Shows past transfers with fees, clubs, and dates
- **getPlayerSocialMedia**: Extracts player's social media links (e.g., Instagram, Twitter)

5.16.3 Integration in the System

These endpoints are called on-demand from the **PlayerView.vue** and other detail pages, providing extra information for the player's profile to make the player profile page as detailed as possible.

6. Testing and evaluation

This section presents the evaluation of the final models developed during this project. Evaluation was conducted by comparing the model predictions against actual market values of players, using both performance metrics and feature importance analysis.

6.1 Linear and Random Forrest Model Evaluation Results

This subsection focuses on the performance of the ensemble model built using a combination of **Linear Regression**, **Random Forest**, and **Gradient Boosting Regressors**. It will show case the final results and show the most important features for the model.

6.1.1 Feature Importance Analysis

The following graph and table outline the most influential features in the Random Forest-based ensemble model, as visually estimated from feature importance charts. These features offer insight into the factors most strongly affecting player market value.

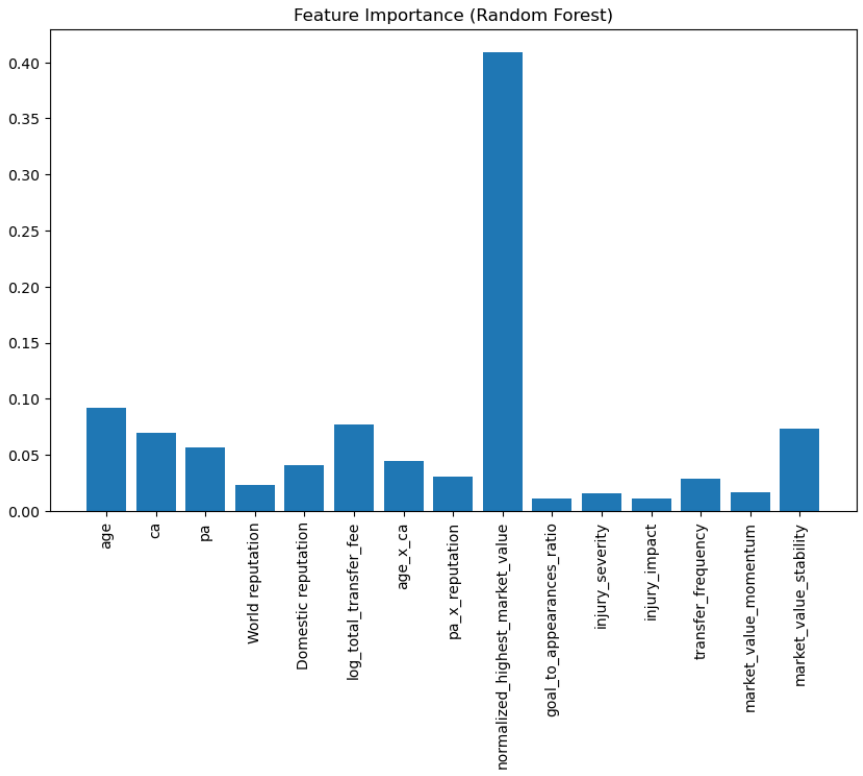


Figure 4 Feature Importance (Linear&RFT)

Feature	Importance (Visual Estimation)	Brief Description
age	~0.09	Player's age — affects peak value and decline phase.
ca	~0.07	Current ability rating — player's present skill.
pa	~0.04	Potential ability — future development capacity.
World reputation	~0.03	International fame — influences market appeal.
Domestic reputation	~0.02	Reputation within domestic league — impacts local demand.
log_total_transfer_fee	~0.08	Player's historic transfer cost (log scaled) — indicates past market valuations.
age_x_ca	~0.04	Interaction of age and ability — captures player performance relative to age.
pa_x_reputation	~0.03	Interaction of potential and reputation — highlights promising well-known players.
normalized_highest_market_value	~0.41	Most important: Highest ever value normalized by age — key indicator of proven worth.
goal_to_appearances_ratio	~0.02	Scoring efficiency — important for attackers.
injury_severity	~0.01	Severity of past injuries — impacts longevity.
injury_impact	~0.01	How injuries affected playing time — affects reliability.
transfer_frequency	~0.02	Number of transfers over career — stability and demand.
market_value_momentum	~0.02	Recent change in value — trend indicator.
market_value_stability	~0.07	Long-term value consistency — measures reliability of market value over time.

6.1.2 Interpretation of Evaluation Metrics

To assess the performance of the ensemble model combining Linear Regression, Random Forest, and Gradient Boosting, four core evaluation metrics were used: **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, **R-squared (R^2)**, and **Mean Absolute Percentage Error (MAPE)**. Each provides unique insights into how well the model predicts player market values.

6.1.2.1 Mean Absolute Error (MAE)

MAE = 3,095,660.94 EUR

This represents the average absolute difference between predicted and actual market values. On average, the ensemble model's predictions deviate by approximately **€3.1 million** from the true player values. This metric is intuitive and less sensitive to outliers, making it useful for understanding general predictive accuracy.

6.1.2.2 Root Mean Squared Error (RMSE)

RMSE = 6,732,595.77 EUR

RMSE penalizes larger errors more heavily than MAE. A higher RMSE compared to MAE suggests the model makes some significantly large prediction errors. An RMSE of around **€6.7 million** indicates the presence of a few players whose values were more difficult for the model to predict accurately, possibly due to exceptional traits or irregular careers.

6.1.2.3 R-squared (R^2)

R^2 = 0.878

This indicates that approximately **87.8% of the variance** in the target market values is explained by the model's input features. A high R^2 value like this shows that the model effectively captures the key drivers of market value across the player population, although it may still miss finer-grained or context-specific nuances.

6.1.2.4 Mean Absolute Percentage Error (MAPE)

MAPE = 48.92%

MAPE reflects the average percentage error between the predicted and actual values. accuracy across players with widely different market values.Mape close 49% indicates that the model performs well overall especially for the young future stars or the aging players.

6.2 XGboost Model Evaluation Results

this subsection evaluates the final XGBoost model developed using an advanced set of e features. The model was tuned using hyperparameter optimization and further improved through blending with other regressors.

6.2.1 Feature Importance Analysis

The feature importance values below reflect the approximate weight of each variable in the XGboost model. This highlights which inputs were most influential in predicting a player's market value

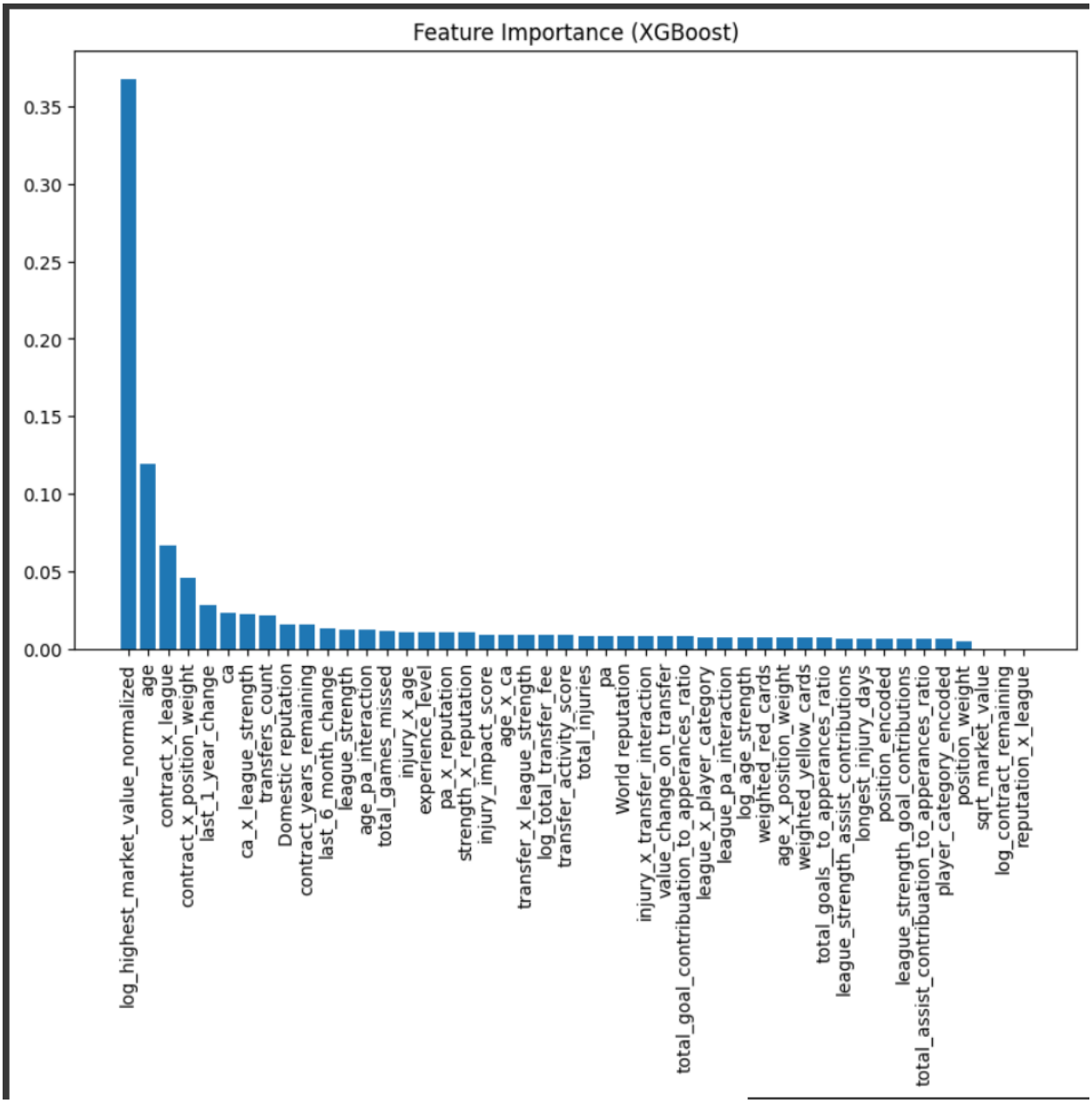


Figure 5 Feature Importance (XGBoost)

Feature	Approx. Importance	Description
log_highest_market_value_normalized	~0.36	Normalized log of the player's peak value — strongest predictor.
age	~0.07	Player’s current age — affects valuation trajectory.
ca	~0.06	Current ability score from FM — present performance.
contract_x_league	~0.05	Interaction of contract years and league strength.
contract_x_position_weight	~0.045	Contract length factored by position importance.
pa	~0.04	Potential ability — future growth.
height_in_cm	~0.035	Player height — relevant for certain positions.
contract_years_remaining	~0.03	Time left on current contract — affects player value.
league_strength	~0.025	Difficulty/tier of the player's current league.
ca_x_league_strength	~0.02	Combines skill and league strength.
Domestic reputation	~0.02	Local reputation — affects visibility.
total_appearances	~0.018	Total career appearances — experience.
last_6_month_change	~0.017	Short-term value change — trend signal.
log_total_transfer_fee	~0.016	Historical transfer fees that were paid for the player
goal_to_appearance_ratio	~0.015	The number of goals scored divided the number of appearances.

experience_level	~0.014	Age × contract years — maturity and commitment.
transfer_x_league_strength	~0.013	Transfer activity in context of league level.
injury_impact_score	~0.012	Impact of injuries on playtime.
strength_x_reputation	~0.011	Influence of fame in competitive leagues.
injury_x_transfer_interaction	~0.01	Combined penalty for injury and movement.
(All other features)	≤0.01	Minor contributors — refined by ensemble learning.

6.2.2 Residual Analysis

To evaluate the prediction quality of the ensemble model, residual plots were generated.

6.2.2.1 Residuals vs Predicted Market Value (Figure 6)

This scatter plot visualizes the residuals (errors) between the actual and predicted market values against predicted values. Most points cluster around the horizontal red line at zero, indicating that the model's predictions are generally accurate.

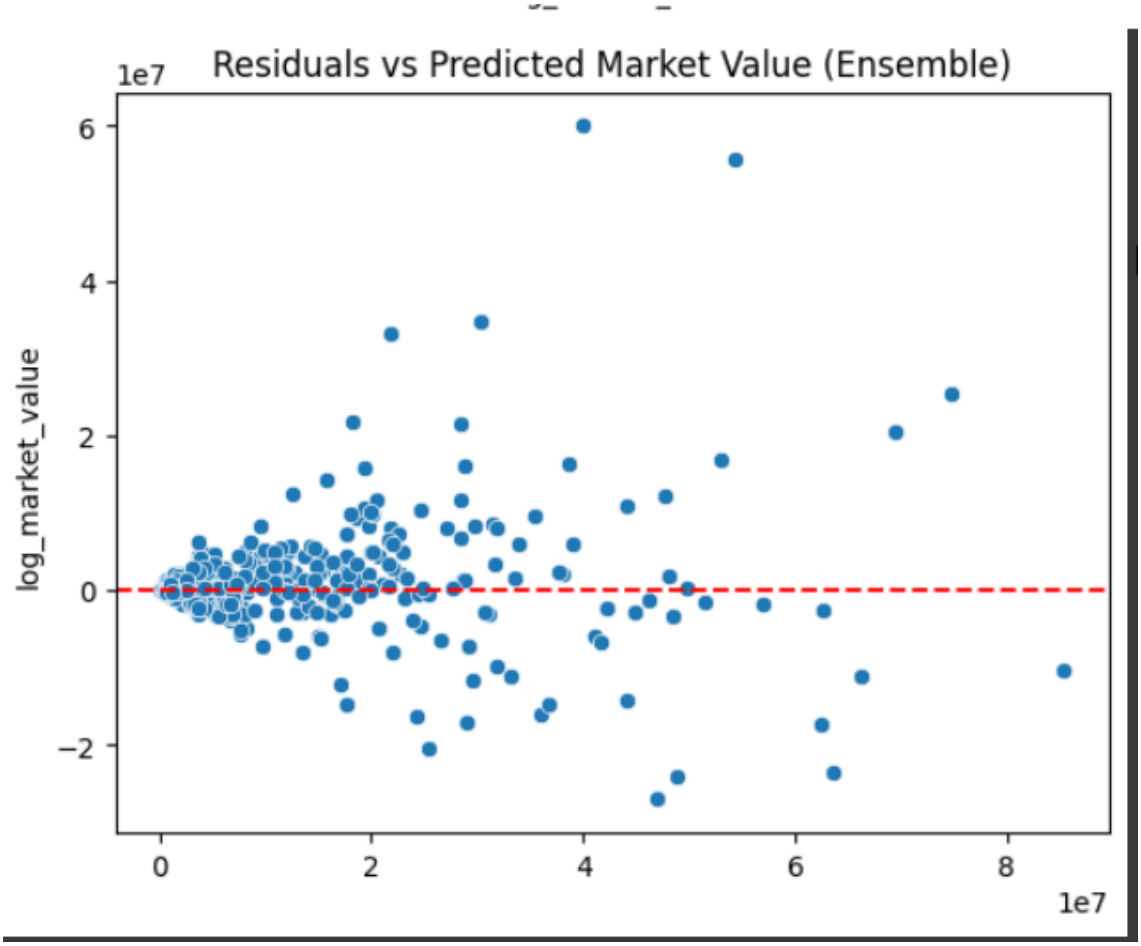


Figure 6 Residuals vs Predicted Market Value (Ensemble)

6.2.2.1 Residual Distribution Plot (Figure 7)

This histogram with a kernel density estimate (KDE) shows the distribution of residuals from the ensemble model. The centered and narrow peak near zero indicates that the majority of prediction errors are small, confirming that the model performs reliably for most players. The long tails represent a few large deviations, which may be due to exceptional or irregular player profiles.

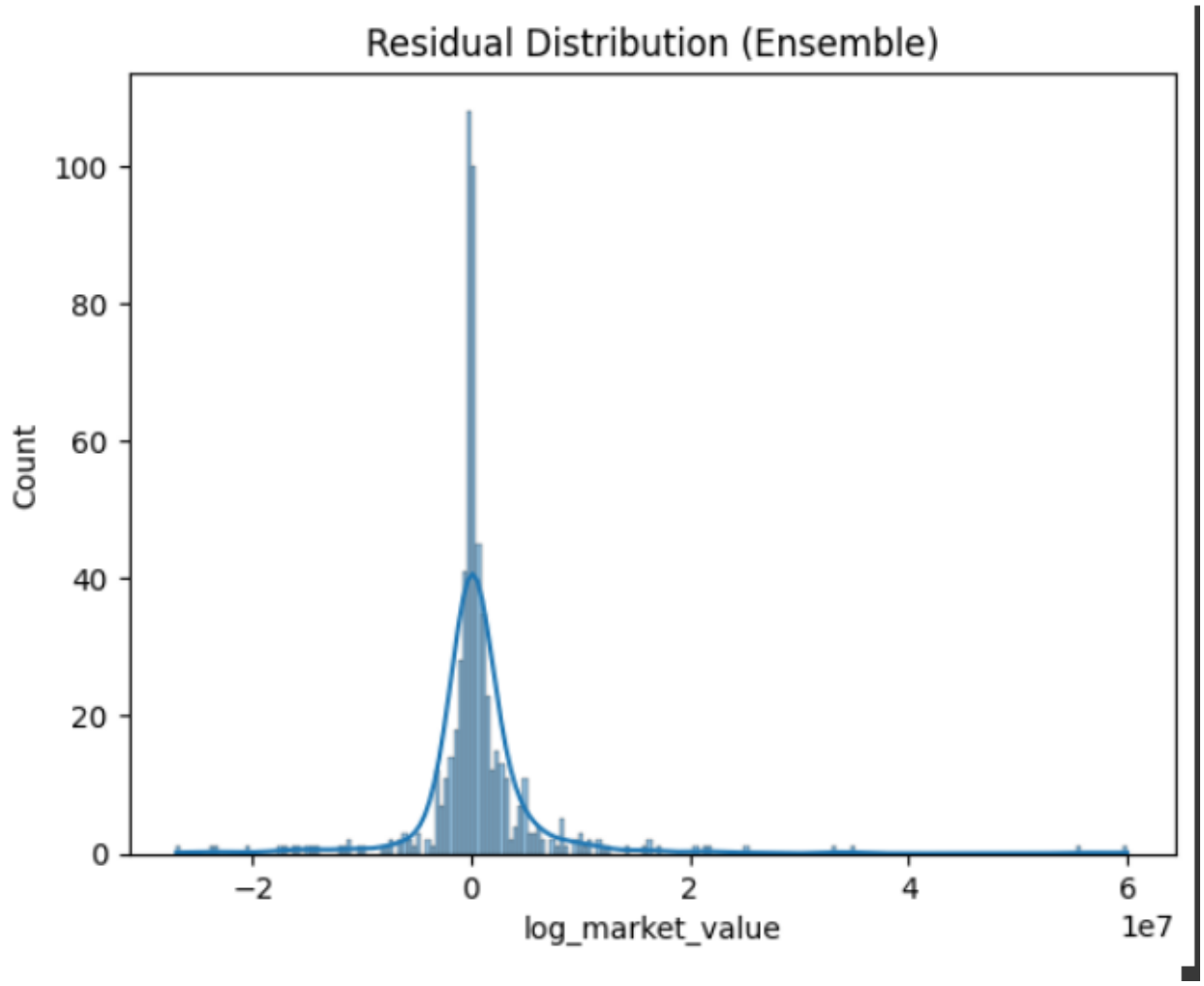


Figure 7 Residual Distribution (Ensemble)

6.2.3 Interpretation of Evaluation Metrics

The performance of the XGBoost model was assessed using four key evaluation metrics: **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, **R-squared (R^2)**, and **Mean Absolute Percentage Error (MAPE)**. These metrics provide insight into the accuracy, reliability, and generalization capability of the model.

6.2.3.1 Mean Absolute Error (MAE)

MAE = 2,614,963.72 EUR

This value indicates the average absolute difference between the predicted and actual market values. On average, the model's predictions deviate from the true values by approximately **€2.6 million**, reflecting a strong level of precision across the dataset.

6.2.3.2 Root Mean Squared Error (RMSE)

RMSE = 5,978,582.45 EUR

RMSE accounts for the magnitude of prediction errors by penalizing larger deviations more heavily. A value of just under **€6 million** suggests that while the model generally predicts well, a few extreme cases with larger errors are present, which is typical when modeling diverse player profiles.

6.2.3.3 R-squared (R^2)

R^2 = 0.905

This means that **90.5% of the variation** in actual market values is explained by the model's input features. This is an indication of a very good model fit and confirms that the features selected effectively capture the underlying factors influencing player market value.

6.2.3.4 Mean Absolute Percentage Error (MAPE)

MAPE = 41.47%

This metric reflects the average percentage difference between predicted and actual values. A MAPE of **41.47%** indicates that the model is generally reliable in estimating values but that relative error can be higher for certain players, particularly those with highly volatile or unusual valuation profiles.

6.3 Comparing Between the two final Models

In this section the evaluation of both the XGboost and the other model that uses both the random forest tree and linear regression will be compared

Evaluation unit	XGboost	Random Forrest tree and linear regression model
Ensemble MAE	2614963.723609162	3,095,660.94
Ensemble RMSE	5978582.450982708	6,732,595.77
Ensemble R-squared	0.905	0.878
Ensemble MAPE	41.47%	48.92 %

Conclusion

The comparison shows that the XGBoost model outperforms the Random Forest and Linear Regression ensemble across all evaluation metrics. The XGBoost model shows better fit to the data and greater predictive accuracy with reduced error values (MAE, RMSE, MAPE) and higher R-squared score. This emphasizes how well advanced feature engineering and gradient boosting capture the intricate trends impacting football player market value.

7. Results and Discussions

In this section I tried to recreate the XGboost from Nikolaos' [6] paper, since he used different dataset from mine, he used combination of FIFA dataset and transfer dataset unlike mine and he didn't provide the code I had to recreate the model from the description and to replace his features with features that exists in my dataset. I also tried to recreate a model similar to my model that uses both the random forest tree and linear but since it wasn't used in the paper and choose algorithm similar to the **LightGBM.**

7.1 XGboost recreation

To replicate the XGboost model referenced by Nikolaos, I followed a multi-step process inspired by the methodology discussed in the original paper. However, I modified the input features based on data availability.

Where Nikolaos combined video-game-based skill metrics and real-life performance statistics, I instead utilized a minimalist feature set containing only:

- the player's **position**,
- their **age group** (grouped into Under 21, 21–25, 26–30, and Over 30),
- and their **highest recorded market value**.

These were used to predict the player's **current market value** (target variable), aligning with the real-world objective of assessing whether a player has retained, gained, or lost value over time.

The dataset was split across different combinations of age groups and positions, and an XGboost regressor was trained for each subgroup. The target variable was log-transformed to normalize skewness in price distribution, following the approach in the reference study. Predictions were then inverse-transformed to evaluate in real currency scale.

7.2 Linear and Random Forest tree recreation using **LightGBM**

To explore a model that approximates a combination of linear regression and decision-tree ensembles (Random Forest), I implemented a LightGBM-based regression model using the same minimal feature set. LightGBM gradient boosting architecture offers tree ensemble power similar to Random Forest while preserving the ability to approximate linear patterns through shallow depth and boosted learning steps.

The feature set was kept identical—position, age_group, and highest_market_value_in_eur—to ensure comparability with the XGboost implementation and isolate the effect of model architecture.

7.3 Evaluation Results

To compare the effectiveness of my XGboost model I will compare the results of both the recreated model from the paper and my XGboost

Evaluation Metric	Nikolaos ' XGboost (Recreated)	My XGboost model
R-squared (R^2)	0.4353	0.9050
RMSE	7,380,815.98	5,978,582.45
MAE	3,971,256.90	2,614,963.72
MAPE	75.64 %	41.47 %

Figure 8 Comparison between paper model and my model

The recreated model used a limited feature set (age group, position, and highest market value), while the final model incorporated detailed attributes such as injury records, contribution ratios, and transfer history. This resulted in a significantly higher R^2 score and much lower error margins in the final model, validating the importance of comprehensive feature engineering in predictive tasks.

Now to compare between my model that uses the linear regression and random forest tree with the **LightGBM** model from the Nikolaos' [6] paper.

Evaluation Metric	Nikolaos's LightGBM (Recreated)	My random forest & linear regression model
R-squared (R^2)	0.3295	0.8782
RMSE	8518275.98	6732595.766127765
MAE	4535321.77	3095660.93
MAPE	85.82%	48.92%

7.4 Discussions

The goal of this section was to evaluate how well simplified versions of the models described in Nikolaos' paper [6] could be replicated and how they compare to the enhanced models developed in this project. Despite the absence of source code and differences in dataset structure, I attempted to reconstruct both my models as similar to mine as possible.

The recreated **XGBoost model** relied on only three core features: player position, age group, and highest historical market value. While this approach captured broad patterns in market trends, the limited feature set resulted in modest performance. The model's R^2 of 0.4353 and MAE exceeding €3.9 million indicate that it struggled to accurately capture market valuation dynamics beyond basic demographic factors.

In contrast, my full-feature **XGBoost model**, which integrated a wide variety of attributes such as injury history, game performance ratios, and transfer activity, delivered a significantly better R^2 of 0.9050. This clearly demonstrates the value of deep feature engineering and domain-specific data in improving prediction accuracy.

To further benchmark model architecture, I recreated a **LightGBM-based regression model** as a proxy for Nikolaos' LightGBM reference and compared it to my ensemble model that combines **Random Forest and Linear Regression**. Both used the same minimal input features for a fair comparison.

The LightGBM model achieved even lower performance, with an R^2 of only 0.3295 and a high MAPE of 85.82%. Meanwhile, my combined **Random Forest + Linear model** reached an R^2 of 0.8782, demonstrating that even with basic features, structural improvements in the model's design can lead to substantial gains in predictive accuracy.

Across both comparisons, it is evident that:

- **Model structure matters**: Ensemble methods that blend linear and tree-based learning yield more stable and interpretable predictions.
- **Feature richness is key**: The most substantial performance improvements came not from the model type alone but from the inclusion of diverse and relevant player attributes.

These experiments not only validate the models developed in this project but also highlight how thoughtful data preprocessing and feature selection can elevate even common algorithms to outperform more complex but under-informed models.

8. Conclusions and Changes

This section summarizes the key outcomes of the project and highlights the progress made. It also discusses the adjustments implemented during development and areas identified for improvement.

8.1 Conclusion

This project successfully developed a comprehensive system for predicting football player market values using machine learning techniques and real-world data sources. Through careful data collection, preprocessing, and feature engineering, multiple models were trained and evaluated, including ensemble approaches (Random Forest + Linear Regression) and XGBoost algorithms.

The final XGBoost model, enriched with detailed player attributes such as injury history, transfer activity, contribution ratios, and reputation factors, achieved an R^2 score of 0.9050 — a significant improvement over recreated minimalist models based on previous academic works. This highlights the critical importance of deep feature engineering and domain knowledge in creating effective predictive systems for the football transfer market.

In addition to model development, a complete web application was implemented using Vue.js and Node.js, with a MongoDB backend. The application not only predicts market values but also offers tools such as a customizable squad builder, player value generator based on player similarity, live match information, and article publishing features.

Overall, the project demonstrates how machine learning combined with thoughtful system design can provide football clubs, analysts, and fans with more objective and insightful decision-making tools.

8.2 Changes

- **Expanded Feature Engineering:**
  Originally intended to use basic player data, the project evolved to more complex features analysis and add using Transfermarkt API new features such as injury impact scores, transfer volatility, and goal contribution ratios, which greatly enhanced model accuracy.
- **Adoption of Ensemble Models:**
  While the initial focus was primarily on XGBoost, ensemble models combining Random Forest, Gradient Boosting, and Linear Regression were also implemented to test different approaches and boost performance.
- **Deployment Adjustments:**
  The backend API was hosted on Railway instead of traditional local hosting to ensure higher availability, scalability, and easier integration with the Vue.js frontend.
- **Additional Functionalities:**
  New features like a Reddit word cloud generator, ROI calculator for transfers, Squad builder, and player valuation module using KNN were added to improve the user experience.

9. Future Work

Although this project successfully created a working system for estimating football player market values, several directions for future developments and extensions exist:

Richer Data Source Access:
The project currently relies mainly on publicly accessible Transfermarkt and Football Manager data. Future research could focus on obtaining more comprehensive and proprietary datasets, including StatBomb, Opta, or other paid sports data services, to further enhance model accuracy and predictive power. These sources offer detailed in-game event data, such as passes completed, xG metrics, and defensive actions, which could significantly improve the sophistication of player value models. A study by Decroos et al. (2020) showed how event stream data from providers like Opta can significantly improve predictive modeling in football analytics [23].

Integration of Live Match Data:
Although some live match data was incorporated via public APIs, full integration of real-time score updates—including live player ratings, in-match injuries, and tactical shifts—requires paid access to advanced APIs. Integrating live feeds from providers such as SportRadar or API-Football Premium could be explored in future work to improve the platform's ability to offer dynamic and real-time player valuation updates. Moniz et al. (2022) have shown that live match data integration can drastically enhance model responsiveness and decision support in sports analytics platforms [24].**Also could have Explored Advanced Machine Learning Models** while Random Forest, Linear Regression, and XGBoost were effectively used in this project, future improvements could involve investigating alternative and more sophisticated models, using CatBoost could better capture player attributes such as position, nationality, and league competitiveness with minimal manual preprocessing [25]. TabNet architectures, could help discover complex non-linear relationships in

large, high-dimensional football datasets, improving predictive performance without requiring heavy feature engineering [26].

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